Imperial College London

AERO96005 - GROUP DESIGN PROJECT

Wholly Reusable Launch System (Recovery)

TRANSMISSION DESIGN FOR A NOVEL SYNCHROPTER CONCEPT FOR THE AERIAL-RECOVERY OF FALLING ROCKET STAGES

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Abstract

The Preliminary Design of *SynchroCapture*, a Novel Ship-Born Unmanned Synchropter for the Aerial-Recovery of Multiple Low-Earth-Orbit (LEO) Rocket Stages using Remote Helicopter Piloting and a Parafoil, is conducted. The design of the transmission set for this novel Synchropter concept is presented - detailing the design of both the **three-stage bevel/bevel/planetary gear set** for power output reduction from a **1600hp turboshaft** engine, at **12000 rpm**, to two off separate rotors, operating at **487.5 rpm**, and the manipulation of this output at the rotor hub by using **dual swashplates**.

This third year Group Design Project (GDP) aims to develop a preliminary design for the mid-air recovery aspect of a wholly reusable launch system, *Recoveroo*, for which as much of the launch vehicle as possible can be recovered to be reused for subsequent launches of LEO satellites. This report forms a subset of the marking criteria for the *AERO 96005 - Group Design Project* module, seeking to test engineering judgement and design trade-offs within large group settings. Whilst the work presented is intended to have a *stand alone* style, numerous other reports in this series concerning *SynchroCaptures'* other components are cited and the reader is encouraged to peruse these also to fully appreciate the highly coupled design process conducted. Table 1 summarises the main vehicle variables, and is provided to give context to the transmission and swashplate designs presented.

Table 1: *Summary of Key the Specifications for SynchroCapture*

Parameter	Value (Metric)	Value (Imperial)
Global Mass & Power		
Weight with Payload (MTOW)	2907 kg	6408 lb
Empty Weight, Helicopter	1500 kg	3307 lb
Fuel Capacity	500 kg	1102 lb
Payload Mass	1400kg	3086 lb
Power Required (Ceiling, ISA)	1193.12 kW	1600 hp
Rotor (For one Rotor)		
Number of Blades	2	2
Blade Mass (each per blade)	40 kg	88 lb
Rotor Diameter	8.05 m	26.4 ft
Rotor Aspect Ratio (Æ)	14	14
Rotor Solidity	0.0909	0.0909
Rotor Disk Loading	45.9 kg/m^2	9.4 lb/ft^2
Rotor Blade Loading	503 kg/m^2	103 lb/ft^2
Rotor Hover Tip Speed	205.4 m/s	673.9 ft/s ($\approx$0.65 Mach at 20000ft)
Rotor Cruise Tip Speed	268.529 m/s	881 ft/s
Rotational Speeds		
Engine Output Shaft	1256.6 rad/s	12000 RPM
Rotor Speed	51.1 rad/s	487.5 RPM
Travel Speeds		
Cruise Speed	185 km/h, 51 m/s	99 kts
Cruise Speed (Loaded)	100 km/h, 27 m/s	54 kts
Max Speed (Tip Mach = 0.85)	225 km/h, 62 m/s	121 kts
Never Exceed Speed, v_{ne} (Tip Mach = 0.92)	243 km/h, 67 m/s	131 kts
Speed (Supersonic Flow, Tip Mach = 1)	264 km/h, 73 m/s	143 kts
Capture Velocity	144 km/h, 40 m/s	77 kts
Timings		
Total Mission Time	1 hr 49 mins	1 hr 49 mins

Acknowledgements

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WRR Team Members

Table 2: *Team Members and Roles*†

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra **Atri Sharma** Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz **Sam Jenner** Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
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†Sub-team leads are shown in **bold**

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1 Introduction

1.1 Motivation & Task Overview

Spaceflight is a historically prohibitively expensive business with the FAA publishing that the NASA average cost for a Low Earth Orbit (LEO) is some $ 23,750/kg [10]. Disruptive innovation from the likes of Elon Musk's *SpaceX* and New Zealands' *RocketLabs* have opened new avenues in which to make access to space not only more economic (decreasing the cost penalty to a forecasted $ 6,078/kg) but also more ecologically compatible, by fully reusing aspects of their launch vehicles. These companies have brought to life the vision on the self-landing rocket stages and trials of mid-air retrieval, in contrast to traditional burn-ups or graveyard orbits; the later of which only serves to increase the problem of so called 'Space Junk'. Mid air capture is nothing new in rocket science however, as in the 1960's the USA performed the first wide-scale retrievals of camera film parachuted back to earth from orbiting spy satellites as part of the CIA's *Corona* missions.

This report is part of a series outlining the design for a competitive and innovative design for a Wholly Reusable Launch System, *Recoveroo*, which consists of both a launch vehicle concept for the transfer of a 160kg satellite to LEO and a related recovery vehicle, *SynchroCapture*, designed to aerially recover multiple descending rocket stages for ship-borne transport and later reuse.

1.2 Structure of Report

Due to the nature of this project a discourse is first held on the ideation of the conceptual design for the aerial-capture vehicle. Occupying the first week of the design period this culminated in the proposal of a purpose built synchropter - a helicopter with **intermeshing blades** - for the interception the descending rocket stages retarded by **parafoils**. The main body of work outlines the preliminary design of a fully-furnished transmission system for this synchropter; consisting of the gearbox assembly and power transfer linkages[1], accessories, casing and oil system as well as the swashplate assembly that connects the transmission to the rotor hubs. A brief overview of features such as the transmission Health & Usage Management System (HUMS) is also hosted[2].

Conceptual Design

2 Recapture Concept and Baseline Ideation

2.1 Feasibility Study of a Purpose-Built Helicopter Concept for Aerial Recovery of Multiple Rocket Stages

Historical air retrievals originally focused on using propeller aircraft, which offer good range and forward speed [12], with more modern attempts favouring the agility and instantaneous strafing capacity offered by rotorcraft (mainly modified commercial brought helicopters). As part of the conceptual design it was thought necessary to evaluate three potential avenues for design exploration: An aircraft concept, A VTOL proposal and a novel helicopter, each of which had their

[1] A breakdown of the terminology used and an overview of the principle of gears along with a summary of those featured in this report is provided in Appendix A.2. It is highly advised that this Appendix is read prior to the analysis presented in Section 3.1 if the reader is not familiar with gears or would desire a refresh on the topic

[2] A full breakdown of the HUMS and maintenance schedules is featured in the related Sub-Systems Report [11].

benefits and detriments explored by a multi-disciplinary teams [13]. Early on in this comparison it became clear that due to **high technological maturity** and the performance in manoeuvrability and lower costs that the **helicopter design was to be the best route** for further evaluation. Among other factors the potential capture mechanisms that could be deployed showed great promise and a high comparative likelihood of success [14].

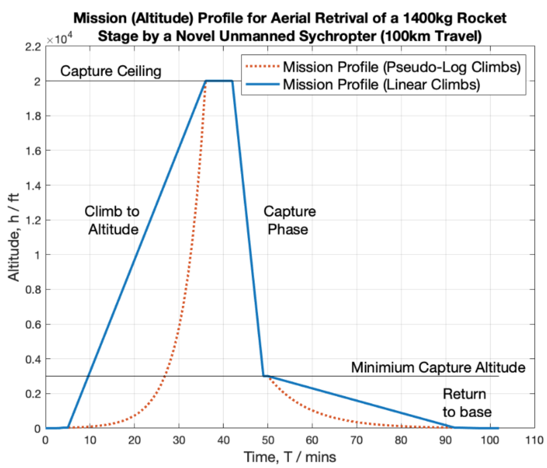
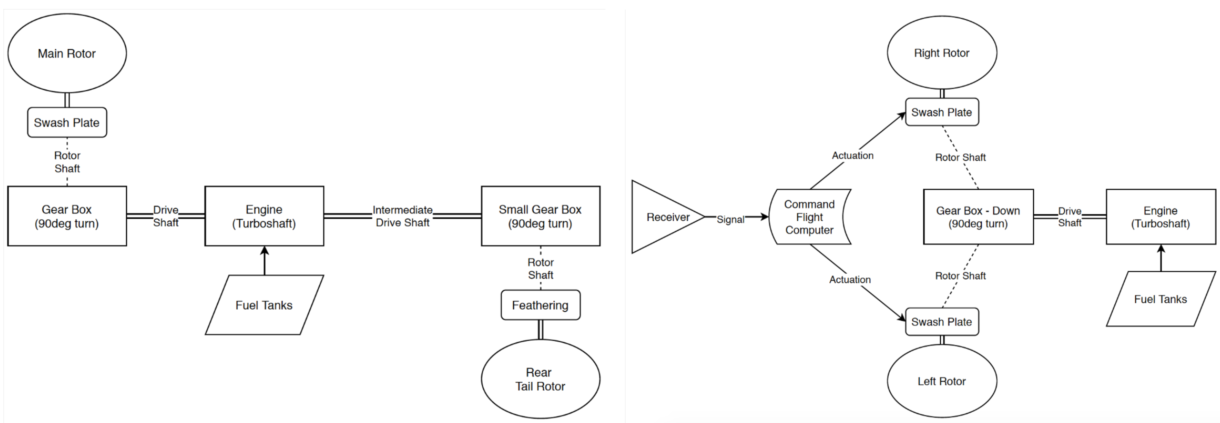


Figure 1: *Capture Mission Profile*

When evaluating the helicopter features such as the global layout and rotor configuration, technological readiness of sub-systems and the design trade-offs of different tail control options were considered. Initial sizing was conducted by contrasting helicopters of comparable roles and configurations. Trade-offs with Launcher design team were also needed for the analysis of capture-instance items (parafoil design). Rotorcraft flight envelope limitations were also a realisation of this stage resulting in operational constraints to avoid unstable operational regions. Namely the necessitation of a forwards speed greater than 40 kts to avoid the Vortex-Ring State (VRS) when looking to achieve any meaningful decent rate to match the rocket stages' descent trajectory.

2.2 Initial Rotor & Thrust Sizing for Aerial-Recovery Helicopter Concept



(a) *Conventional Layout (SMR)* **(b)** *Synchropter Layout*

Figure 2: *Block Diagrams of a Synchropter Transmission presenting a simpler transmission footprint, not requiring tail rotor transmission links when contrasted with that of a Single Main Rotor (SMR) design*

An intermeshing configuration was decided on[3] for the main rotors; maximising the lifting efficiency[16] that could be obtained for the carriage of a large cargo while allowing for a compact lightweight airframe and eliminating the need for an anti-torque device. It should be noted though that an additional rotor pylon and hub was needed and required careful design to minimise this drag penalty [17]. At this stage that the first idea of the mechanical transmission and directional control systems for *SynchroCapture* was realised, as given a rotor configuration, baseline sketching could be conducted.

[3] A full trade-off discourse may be found in the relevant Aerodynamic Optimisation Report [15]

After evaluation of the global design trade-offs (In particular those of a coaxial design vs a synchropter) an initial rotor sizing [18] and engine sizing took place [19] using combined momentum theory approach and also a altitude **power-lapse** featuring approach [20]. This stage mainly consisted of evaluating the power that needed to be produced by the engine to attain hover at the maximal ceiling of 20,000ft. Due to an overestimate in the predicted cargo mass to be captures (Potentially up to 4 tonnes at this stage) the first engine estimates were grossly overpowered, being in the region of 2,400hp. The need for such a large engine was (in addition to the slung load carriage) to compensate for the 50.3% engine-power[4] and lifting efficiency reduction at altitude.

Preliminary Design

3 Design of a Synchropter Transmission System

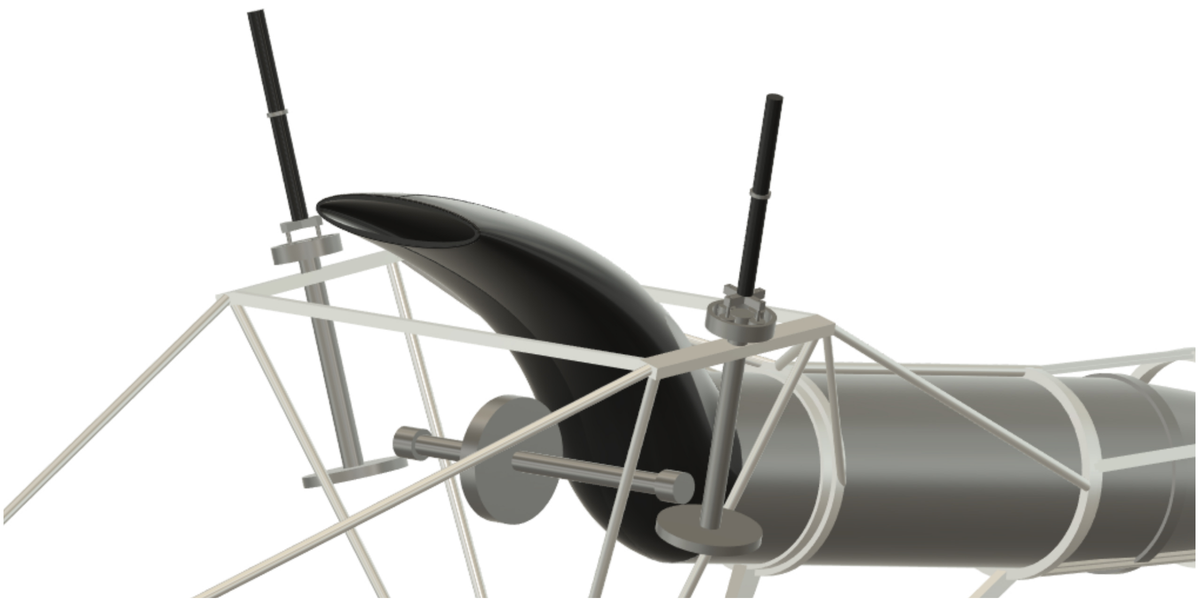


Figure 3: *Final transmission design, rendered as constant running pitch circles [1], with rotors removed. The 'S-Duct' of the engine can clearly be seen in between the pylons. Note the compact planetary gears located halfway along the rotor shafts. (The right rotor rotates clockwise when viewed from above. See Appendix A.1)*

3.1 Gearbox Overview

The transmission system transfers the fixed 12000 RPM shaft output from the single 1600hp two stage turboshaft engine [19] to two rotors, greatly reducing the revolutions per minute. The output is split along two shafts of 28° included angle and end separation 5ft to provide sufficient clearance and prevent blade hub collision at maximum deflection (3.5g load case [21]). The transmission system also provides a source for any electrical[5] power generation. Contrary from more conventional helicopter designs there is an absence of a tail rotor (the two rotor blades rotate in opposite directions) and hence no tail gearbox or shaft linkages in the eppenage. Consequently the gearbox

[4] A function of the density ratio at 20,00ft and how this affects lift and engine aspiration (See Appendix A.6)

[5] In *SynchroCapture* all non-cooling hydraulic systems i.e. brakes and actuators have been replaced with more modern, simpler, compact and lightweight electro-hydrostatic systems [22]

layout is simplified, despite of the need to supply power to an additional main rotor. As a result of the conceptual formulation airing more on the side of **current and near future deliverable technologies** a mechanical transmission is designed instead of a more novel electromagnetic transmission [23] which in future years will hopefully reduce the mechanical wear and improve reliability.

Table 3: *Gearbox operating values at normal continuous operating conditions*

Gear Set	**Reduction Ratio** / [Dimensionless]	**Input & Output Speed** / RPM	**Efficiency** / %
Bevel (Drive) Set	1.875:1	12000:6400	97.0
Bevel (Face) Set	3.571:1	6400:1792	97.0
Planetary Set	3.333:1	1792:538	99.5
Total Transmission	24.3:1	12000:538	93.6

3.1.1 Design Drivers

Drive systems can be a large fraction of a helicopters MTOW (usually some 10%) due to the need to transform a large shaft power into useful lower speed rotation[6]. Weight increases proportional1y to torque - thus it is desirable to have a transmission system has low torque (high RPM) carried through as much of the transmission as possible. For a helicopter issues surrounding weight are obvious: to generate sufficient force to lift a larger mass, a helicopter requires a more powerful engine and larger rotor blades, increasing drive system weight - in turn requiring more lift! [24], hence the design driver is to **gain maximal efficiency for the lightest weight gearbox** that transforms the power required. The following assumptions are made for this Preliminary design:

- Ancillary parts (keyways and bearings etc) do not contribute a significantly design challenge to the transmission thus are mentioned only briefly.
- Transmission loads (Torque, T) are a only function of Power, P and RPM ($T = {}^{P}/_{RPM}$). These loads are evaluated in the first linkage, which experiences the most resistive stress.
- Efficiency losses result in a linear drop in output compared to the idea case.

Table 4: *Gear Specifications including Pitch Diameters & Teeth Numbers*

Gear	**Pitch Diameter ($\varnothing_p$)** / mm	**Depth** / mm	**Draft Angle** / °	**Teeth per 100 mm Pitch**	**Number of Teeth**
Drive Gear	160	40	30	35	56
Splitting Gear	105	50	60	35	105
Axle End Gear	70	60	30	40	28
Face Gear	250	25	46	40	100
Sun Gear	60	40	0	35	21
Planet Gear	40	40	0	35	14
Ring Gear	140	40	0	35	49

[6]Rotor speed is a function of the tip's speed and rotor diameter, such that critical Mach is not exceeded. Often with the limitations on increasing rotor radius (which demands a lower tip speed) it is the job of the transmission engineer to cut weight to increase performance.

3.2 Gearbox Design & Architecture

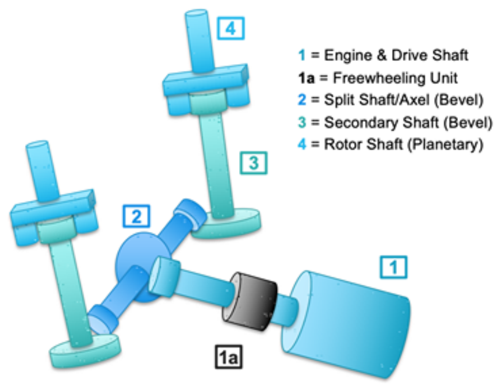


Figure 4: *Number based sub-assembly diagram of the developed transmission system.*

The gearbox developed consists of two bevel gear sets followed by a planetary gear stage. The design is symmetric and yields a total gear reduction ratio of 24.6:1[7] at 93.6% efficiency. Each stage is summarised in Table 3 and rendered in Figures 3 & 4. The engines' free turbine connects directly to the high speed shaft, operating at 12,000RPM when the clutch is fully engaged. The drive shaft is supported by a set of bearings which are attached to the engine casing and further mounted to the airframe trusses. Shaft rotation is anti-clockwise when viewed from the aft end of the engine. This shaft delivers engine torque to the first gear in the transmission, a bevelled pinion, which connects to a splitting gear (1.875:1, 12,000RPM to 6400RPM, 97% efficiency) which is mounted to a high speed axle directing towards the base of the rotor shafts.

Each axle-end terminates at a gear, bevelled to allow for intersection with a face gear (reduction ratio 3.571:1, 6400RPM to 1792RPM, 97% efficiency) at the bottom of a intermediate speed secondary shaft on the same axis of rotation as the rotor blades. The secondary shaft carries the output upwards to the sun gear of a fixed-ring common direction planetary set. Here there is further reduction to the speed desired for the blades (reduction ratio 3.333:1, 1792RPM to 538RPM, 99.5% efficiency). A carrier connects the planets to the low speed rotor shaft - exiting to the swashplate and rotor-blade assembly.

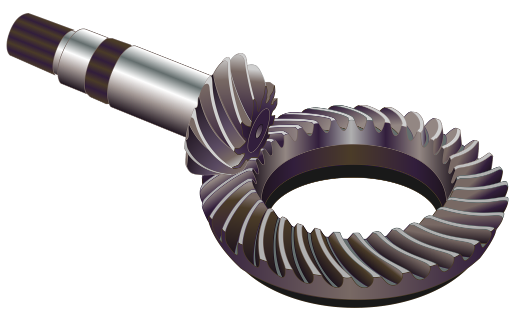


Figure 5: *A spiral bevel set, similar to that joining the axle to the facegear [2]*

3.2.1 Contact Ratio

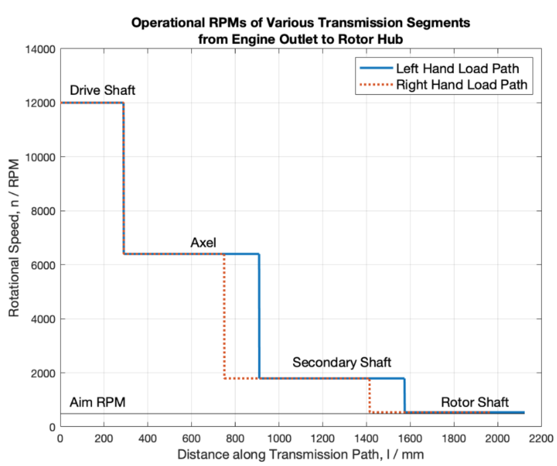


Figure 6: *Plot of the Ideal RPM and relative reductions along the transmission paths.*

The parameters for the gears used are summarised in Table 4, a common teeth spacing was sought to be used reduce the number of reamers used in maintenance. To increase the transfer of mechanical loading and lower the noise produced high contact ratio helical & spiral bevel gears will be used (Figure 6, also Appendix A.2). The use of helical gears also requires the shaft to be supported by thrust bearings, which can resist the axial load caused by the 20° helix angle (a value was chosen as a compromise between improving contact ratio whilst minimising the resultant axial thrust). The use of helical gear teeth along with the gear lubrication system (Section 3.4) are included to improve efficiency.

[7]In theory the reduction value is 22.3:1 i.e. **538RPM** output, however with **mechanical system losses** of RPM (6.4%) this falls to a value much nearer the desired rotor revolutions of **487.5RPM**

3.2.2 Modelling Gearbox Losses

Advancements in the performance of modern helicopter (and automobiles) are testimony to how ratio tuning and gearbox advancements can yield appreciable increases in performance. Dudley's Gear Handbook [4] was used to model the initial gearbox efficiency. Whilst not the most modern text Dudley provides an easy to use insight for those who are not savants in the realm of power transmission. As the reduction ratios in question are all low (less than 5:1) hence as a **first approximation** a standard 97% figure is appropriate; hence an initial total 92.1% efficiency. For the purposes transmission loss in engine sizing 50hp was allotted to cover mechanical losses in the first stage, based on historical regression data for comparable power (using a single planetary and bevel gearset). As the design progressed these estimates could be refined. With the consultation of more modern resources it was found that **helicopter turbo-machinery can obtain extremely high efficiencies**, often above 95% for the entire transmission [25], especially if the right selection of **high grade and high running temperature lubrication** is used alongside the gears. The breakdown of final efficiencies and reduction properties of each stage is displayed in Table 3.

3.3 Shaft Sizing

A series of shafts connect the gears and transmit power from the engines to the rotor. The transmission is constructed to reduce the 12,000RPM input to the rotor RPM at cruise (487.5RPM at 185kph). The shafts are **designed to withstand significant torsional shear**, superimposed with some axial tension stress and bending stress from vibration. It is worth noting that the the shaft assemblies are designed to operate *subcritically* (below self resonant frequencies [26]) to prevent either engine vibration or rotor oscillation from exciting a natural mode[8] within the operational RPM range. The shafting is in four main sections (Figure 4, summarised in Table 5), each of external ∅50mm, excluding the engine output shaft and rotor shafts (See Section 3.3.1).

Table 5: *Summary of Shafting Specifications*

Shaft	**Operational Speed** / RPM	**Diameter (Outer ∅)** / mm	**Wall Thickness** / mm	**Length** / mm	**Material**
Drive	12000	20	*Solid*	250	Titanium
Axle	6400	50	2	780	Steel
Secondary	1792	50	2	500	Steel
Rotor	538	40	2	500	Steel

The first shaft (of total length of 1450m) is diffusion bonded directly to the free turbine stage and has 250mm protruding out of the front of the engine. At a declination of 10° from the horizontal, the shaft is attached via a high strength keyway (Appendix A.2) to the first reduction gear. In contrast to later sections high strength lightweight titanium is used for this shaft due to the turbine and engine operational temperatures. The secondary and rotor shafts share a common axis due to the planetary gear arrangement (and to minimise wear/eliminate the fatigue load that would have been placed on a flex-coupling), sharing the offset with the rotor blade above at a 14° tilt outwards - allowing 28° rotor clearance [21] - and having a 3° forward cant, for stability [28]. The shafts are supported near there ends by bearing blocks attached to the transmission housing that take the self weight of the linkages but allow for free rotation.

[8]Excitation frequencies are usually an integer multiple of a the natural frequency. In a detailed design stage a method such as **Rayleigh-Ritz** can be used to find the first natural mode. The use of these analytical methods and details on subcritical running can be found in FAR29 [27].

3.3.1 Torque Tube Calculations

Transmission shaft sizing was based on Equation 2 to determine[9] the minimum possible solid shaft diameter, d_{Min}, and Equation 3 for wall thickness, t_{Min} and required in the drive segments to resist deformation the engine torque, $T = {}^{P_{\text{Engine}}}/_{RPM} = 952$ Nm. The Torsional constants, J, were determined for each section and a 1.5 times safety factor was applied. A minimum 1.62 mm wall thickness was calculated for the steel members, given their external diameter, $d_o = 50$ mm, which was rounded to 2mm as industrial sizing for ERW (Electric Resistance Welded) tubing skip from 1.5 mm to 2 mm, with the former not meeting requirements.

$$\tau = \frac{Tr}{J}, \qquad J_{\text{Solid}} = \frac{\pi}{2} r^4, \qquad J_{\text{Hollow}} = \frac{\pi}{2}(r_o^4 - r_i^4) \tag{1}$$

$$d_{\text{Min}} = 2 \cdot \left(\frac{T}{\tau \cdot {}^{\pi}/_{2}} \right)^{\frac{1}{3}} \tag{2}$$

$$t_{\text{Min}} = \frac{d_o - d_i}{2} \quad \text{Where} \quad d_i = 2 \cdot \left(\frac{({}^{d_o}/_{2})^4 - T \cdot {}^{d_o}/_{2}}{\tau \cdot {}^{\pi}/_{2}} \right)^{\frac{1}{3}} \tag{3}$$

Material selection set the value of the limiting shear stress, τ, which were 250 Mpa for Steel (used for the majority of the shafts) and 760 Mpa for Titanium[10]. The larger τ of Titanium supported it's use in the drive shaft - as this shaft to runs the length of the engine to connect to the free-turbine at the back (counter-rotating inside the compressor) a **solid bar ∅20 mm** was selected. Both materials see use in modern helicopter transmissions although in future composite shafts may see use and posses the potential for significant weight reduction [6].

3.4 Lubrication System & Gearbox Cooling

SynchroCapture has two main oil systems due to different operating conditions[25] - one for the engine (sescribed in a Sub-Systems Report [29]) and another for the gearbox, located directly in front of and below the Axel. The gear oil system is its own subsystem with a radiator unit, independent piping and numerous ducts. These lines are thermostatically regulated to ensure efficient heat removal. Oil from the radiator is recirculated into usage; being forced through splash-jets, a methodology shown to be superior to vapour lubrication for cooling transmissions with multiple moving parts[8]. Numerous automated and visual oil level gauges surround the gear casing, and a main sump valve for oil draining and replacement is placed on the aft underside of the transmission assembly, for access when the rear panel of the fuselage is removed. As a safety precaution for abort and descent and also to comply with FAR 29.927[11] the gearbox is designed with 30 minutes dry running time, the gear material (6-1 Series high running temperature Steel) assists in achieving this.

The initial weight estimate for cooling oils was 60lbs [24] (although the recirculation system will weight significantly more[29]). It is assumed that in operation the bearings and gears operate in sealed condition (the cavity of the gearbox casing was assumed to be impermeable) with fluid being supplied to the gears via a mesh manifold and jets. Further gear design is key to ensure that lubrication is effective and not immediately is flung off by gear rotation[12].

[9] Code presented in Appendix B. The most constraining was taken for diametric and wall thickness sizing.
[10] These figures do vary with the specific grade of alloy used but were thought sufficient for first pass calculations.
[11] Any failure which results in loss of lubricant will not prevent continued safe operation for at least 30 minutes[27]
[12] With less time for the jet to penetrate between the teeth faster gears also suffer from being less well lubricated[25]

3.5 Accessory Accommodation & Electrical Power

The gearbox and related shafts also provide power to additional rotorcraft systems and features various accessories key for the operation of the vehicle. Accessory modules are situated around the main axle and on the front side of the transmission assembly, close to many of the subsystems in need of power. These modules include a 50kW AC generator used to power avionics in flight situated on the port side of the main axle and a hydraulics group (driver pump and reservoir) for the local gearbox lubrication system. Descriptors of the linked generator and it's power electronics network can be found in the related Sub-Systems Report [22].

3.5.1 Transmission Clutch

As *SynchroCapture* is only single engined there is no need to receive two engine inputs through a spring clutch as is the case in multi-engined helicopters; thus an common axis **centrifugal clutch** can be used. This freewheeling unit is situated between the engine input and the first gear. Being the location of lowest shaft torque mounting a clutch here will allows its weight to be minimised [30]. This unit is not directly attached to the shaft due to issues with dynamic balancing[13] at high speed, but rather to the casing surrounding the transmission assembly (Section 5.1). The unit consists of a clutch that permits the rotor blade assemblies to slowly pick up speed from the engine on start-up to prevent overstressing of mechanical parts and also to auto-rotate without engine drag in the event of engine failure (of if a large mismatch in RPM speeds occurs). The freewheeling unit is situated before the split of power to the rotors, ensuring that the blade phase[14] will not be affected and that the blades will not be allowed to collide - even in autorotation. The clutch is disengaged while the engine is offline, spooling up or powering down [31] and features a transmission lock that prevents the rotors from accidental windmilling when the UAV is offline on the landing deck.

3.5.2 Rotorcraft Brakes

Aft of the clutch is found the 'drive end' braking system (See Figure 25 in Appendix). This brake is in place to serve to slow the transmission upon landing. As a guideline this system must be sufficient to bring the rotor to a stand still within 30 seconds [31] after powerdown - a compromise between minimising inertial stresses on the system and also allowing for rapid access to the helicopter.

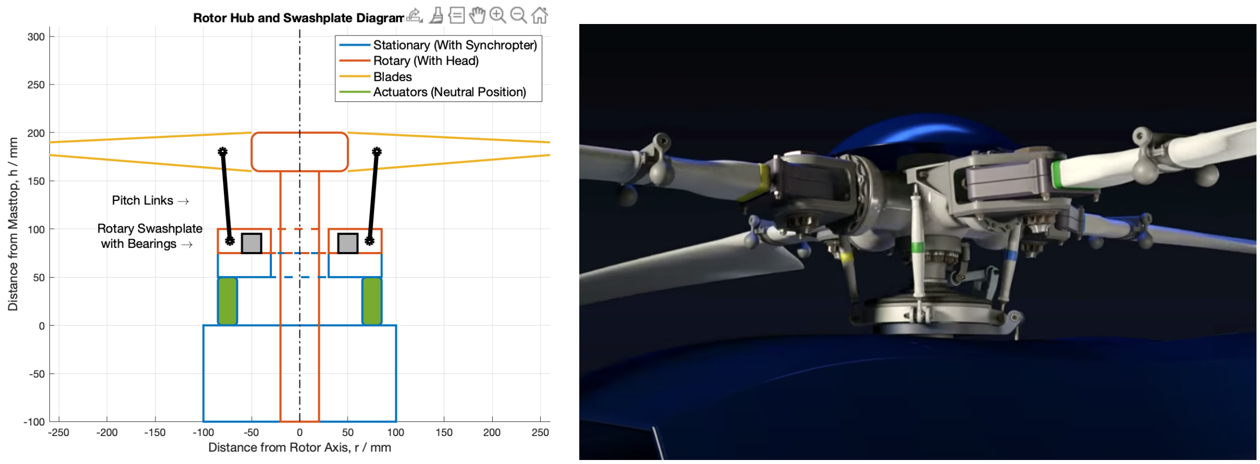
A second set of breaks is located on each of the 'rotor ends' of the transmission near the planetary gear-sets to decrease the risk of rotor overspeeding[15] that could damage the gearbox in high loading scenarios such as the capture manoeuvre' where it is likely that there will be a sudden change in blade pitch to increase lift. With a healthy maintenance schedule this device will reduce the likelihood of over torqued conditions leading to mechanical damage. All brakes are Electro-Hydrostatically actuated [22], eliminating the need for a full hydraulic power unit. The absence of a pilot to provide a manual input to the cockpit brake master cylinder this system is completely fly-by-wire. The brake unit is composed of the actuating circuitry, the brake disc itself and a brake accumulator - which acts as a completely redundant system.

[13]Think of this as the relative stability of a shaft rotating with a large asymmetric mass, the larger offset/eccentricity the less dynamically balanced and the worse the transmitted forces and moments caused by the gyration [26].

[14]The intermeshing configuration demands a constant 90° offset in the positioning of the blades, such that the passing blade to sweep out the area formerly occupied by the opposite rotating blade.

[15]Overspeeding is the unwanted acceleration of the rotor as a result of the aerodynamic forces on the blades flexing them, decreasing the polar moment of inertia, hence like an ice skater drawing their arms in, causing acceleration. This has a chance of feeding back through the gearbox, which is designed to operate at constant RPM.

4 Design of Synchropter Swashplate Sets



(a) *A Matlab realisation of the Swashplate Assembly* **(b)** *Hub of a EC145, which has comparable rotor articulation [32]*

Figure 7: *Hub design (Left), showing clearly the pitch linkages that connect the swashplates to the blades*

A titanium forging, each rotary wing head connects two blades to the rotor shaft via several different linkages which serve to transfer torque from the gearbox into the rotor disc. The head carries the rotor lift load down the rotor shaft to a standpipe structure (Section 5.2) connected to several mounting points on the fuselage as well as transmitting the input of collective and cyclic pitch[16] from the flight controls to the blades via a set of Swashplates and Pitch Links (see Sections 4.3). The Design of the swashplates (and some of that of the rotor hub) are discussed here. The rotor blade kinematics and structural analysis are covered separately in a related Structural Report [21].

Table 6: *Summary of Control Authority Specifications of Swashplate*

Variable	Value	Variable	Value
Maximum Collective / °	24	Response Time / s	0.1
Maximum Cyclic / °	±12	Actuator Stroke Length / mm	36
Pitch Horn Length / mm	85	Actuator Force (ultimate)/ N	3797

A key decision was the type or rotor system that *SynchroCapture* would mount; the usual choice is either a Rigid, Semi-Rigid or Fully-Articulated Design. Despite the systems' agility and responsiveness a rigid rotor (i.e. the blades are rigid) was ruled out due to the need to damp and attenuate at least the flapping motion of the blades, which was a major concern due to the rotors intermeshing and also the high blade fatigue this configuration causes. A **fully articulated head was chosen** due to it's benefits in attenuating lead and lag, however due to project time constraints a large trade-off including more advanced hingeless and bearingless designs was not conducted.

- **Semi Rigid/Teetering:** A flexture based design (for high control authority which may be useful in capture operations) this design allows the blades to feather in pitch and flap but any leading and lagging is absorbed purely by the blade
- **Fully Articulated:** Allows the absorption of stresses and vibrations in all 3 blade axis including from lead/lag via sets of hinges and an additional damper.

[16]It is assumed that at full collective travel there is cyclic. In addition the blades are very close to stalling.

4.1 Lead Lag Dampers

An advantage of fully articulated rotors is a longer fatigue life, this is because instead of the blades oscillatory modes being absorbed by the blade itself they can be passed on to dampers at the hub. One such mode is the leading and lagging of the blade during rotation, which is nullified by a very strong magneto-rheological[17] damper. Due to the inter-meshing nature of the rotors these were considered to be crucial to eliminate the buildup of oscillations in the advancement or delay of the blade. Further it is stipulated in FAR 29 that if the rotors must be phased for intermeshing then within reason a system must be in place to provide "constant and positive phase relationship under any operating condition" [27]. In addition to the setting of this delay by the mounting of the rotorshafts in the gearbox the damping system ensures that regardless of oscillation this offset remains. Other elements of the damping structure and rotor hub structural design may be found in the related blade and hub structural design report [21].

4.2 Swashplates vs Flaps

SynchroCapture a more traditional swashplate design is used for the collective and cyclic pitch inputs; this is in contrast to some synchropters - notably those manufactured by KAMAN - which utilise a unique trailing edge flap system. Much like flaps on a conventional aircraft wing these adjust camber and aerodynamic forces on the blades to produce rotorcraft movement, however can be very difficult to model - especially given this projects' time constraints. The main tradeoffs between these designs are summarised in Table 7. Whilst these flaps will not be used in this case there will however be a small trim tab on the blades which are a standard feature on almost all helicopters [18].

Table 7: *Trade-Off Table between Swashplates & Flaps*

Swashplates	Trailing Edge Flaps
Benefits	
• Mature and understood technology	• Excellent mass reduction capability
• Simple impact on rotor aerodynamics	• Allows for fine blade control
Disadvantages	
• Large drag contribution	• Relatively immature technology
• Mechanical linkages add complexity	• Complex 'in blade' power electronics & Slip-Rings

4.3 Collective/Cyclic Control Authority & Pitch Links

Within the aerial-retrieval mission the capture manoeuvre itself is perhaps the most pressured. Precision control is needed to successfully mate the descending stages parafoil with the helicopters' capture mechanism [14]. Aircraft motion is controlled by tilting the rotor planes (See Figure 21 in Appendix). The swashplate is the mechanical linkage at the top of the mast that transfers control input to output through **adjusting blade pitch** (feathering). This control is usually **collective**, when all of the blades pitch are adjusted for altitude control, or **cyclic** which allows for helicopter strafing, pitching and rolling by adjusting the pitch of blade only in a relevant section of the rotor plane. The synchropter layout, being somewhat different to a conventional SMR, requires careful

[17] In these dampers the viscosity and hence stiffness if the fluid can be changed by applying a magnetic field. This eliminates any mechanical valves and allows for closed-circuit fine oscillation control.

adjustments result in normal multi-axis control[18]: To **roll**, both rotor planes are laterally tilted. To **pitch**, longitudinal tilt is applied (in a common direction). To **yaw**, the rotors are longitudinally tilted in opposite directions.

Situated below the hub on a spherical bearing within the pylon fairing the swashplate takes input from 4 off Electro-Hydrostatically actuated pistons [22] capable of a 36mm stroke length. These pistons can produce 3797N (ultimate[19]) of force each to overcome aerodynamic loads at the pivot to command actuation with a 0.1 second response time (less than one blade rotation). High strength titanium pitch links (also called pitch horns) connect the swashplate to the blades and transfer the commanded manoeuvre. There are 4 links, two on each side of the rotor (See Sectional Figure 7a) of length 98mm which enact a maximum 24° collective or a ±12° cyclic [9]. An separate offset linkage is used in the control mechanism due to the gyrospcpic procession that occurs in spinning rotor blades[20]. Although negative pitch was considered as a method to plant the helicopter onto the deck of a ship in rougher seas it was found that at 0 collective there was already a large downward force as the blades are quite negatively twisted.

5 Structural Integration

5.1 Gearbox Housing

The gearbox and transmission components are enclosed within a 6 part cast aluminium casing. The multi-part clamshell design is envisaged to reduced maintenance time by encouraging flight technicians to remove only the relevant sections of the casing during inspection/repair. The casing attached to the fuselage frame via flanged lips with anti-vibration fasteners (tabs from the frame with rubber washers and nylock nuts), whilst the shafts themselves lie within bearing blocks attached to this casing. Mechanically isolating components reduces transmissibility factors throughout the airframe, increasing fatigue life. A more in depth analysis of the airframe with some detailing of this linkage can be found in the Airframe Structures Report [33]. The housing is designed to provide an element of protection for the transmission from external impacts, providing ballistic protection to internal subsystems from large-energy fragmentea in the event of a (very unlikely) catastrophic gearbox failure and serve as a barrier to oil seepage/spillage from lubricated gears.

5.2 Mast Design & Weight Estimation

The swashplate assembly is enclosed within the aerodynamic fairing of the pylon to reduce the drag contribution from the two rotor hubs and pylon structures (As much as 30% of the total forward drag [20]) whilst also allowing for efficient structural mounting of the rotor to the fuselage and the transmission of lift forces to the airframe. These are tilted outwards for rotor clearance and to prevent downwash pressure buildup. The standpipe structure [33] within the pylon serves as this load transfer member - minimising severe cyclic loads on the transmission housing; allowing the lift loads to be passed into the helicopter's main body through a separate truss structure, taking the place of heavy frames in conventional aircraft. For the MTOW calculations a multi-variant sizing methodology was used [20], giving a total transmission mass of 261.7kg - likely an overestimate but

[18] Unlike a tandem (chinook) configuration there is no rotor collective differential for yawing.

[19] Due to the swashplates importance in the mast as a single point of failure for the helicopter a higher safety factor should be used in subsequent detailed design.

[20] Summarised best: If the offset linkage were not present then the pilot's remote cyclic stick input would have to be 90°out of phase... pushing right to tilt the disk forward, and forward to gain left travel etc.

accommodating cooling, casing and accessories, which were difficult to predict otherwise[21]. From the CAD model presented and MATLAB summation (Appendix B) it was found that at least 72.8 kg of the transmission weight is the metallic mass of the shafting and gears. Additional weight calculations are summarised in the related Weight and Balance Report [29].

5.3 Health & Usage Monitoring System (HUMS)

There are to be sensors at every level of the transmission, these include systems for monitoring oil temperature in the coolant system as well as to detect low oil pressure, which whilst undesirable is not an occurrence that would outright lead to a mission abort due to the dry running system (Section 3.4). Thermal sensors will also be in place to real time report the gearbox temperature and operating speed to the ship-borne command centre. For maintenance logging a chip detector system will also be present to monitor lifetime wear on the gear faces. This device is electromagnetic and detects the concentration of residual ferrous shards from warn[22] gears in the circulating oil.

6 Summary

The work presented outlines the preliminary design for the complete synchropter transmission system, in addition to exploring some elements of the rotor head and swashplate system. The design presented contains many technologies that are currently used in industry or are derivatives of foreseeable future technologies. Overall, the design process has been an interesting and thought-provoking return to the core Mechanical Engineering Syllabus which forms a major component of Aerospace and the conversion of an idea on paper to a physical realised flying machine.

6.1 Suggestions for Future Research

With advances in battery technology and other emerging cutting edge technologies such as rotor morphing, it would be desirable to evaluate possible utilisation of these technologies in the design. In the analysis presented some of these features were considered, but deemed too technologically immature - the **running theme** for the transmission design has been it's **feasibility and technological maturity**.

This said emerging technologies could produce several specific advantages for the rotorcrafts' mission: An all-electric drivetrain would not suffer the altitude power lapse that a more conventional helicopter turboshaft does (which itself was chosen due to being less susceptible to power-lapse than an internal combustion engine). There are promising advances being made in electric power, with several recent electric aircraft conducting short flights [35] & [36] - albeit with a very large mass fraction of batteries, this would have to be resolved before any transition to electric. Likewise filament wound composite in the transmission could afford great weight savings and are advised for consideration in any subsequent detailed gearbox design (together with more advanced gear type selection and evaluation of axial and radial loading). Future designs may also seek to eliminate the swashplates entirely also, favouring Individual Blade Control (IBC) where full length actuators replace the pitch links and swashplates. With advances in actuator speed and some enlargement/strengthening to counter any blade pitching moment this alteration would be very feasible if not for such a system being too large for *SynchroCaptures'* high altitude mission where the weight penalty would be severe.

[21] This value seemed more reasonable than other methodologies [34] which suggested a mass exceeding 300kg.
[22] 'scuffed' or 'scored' teeth in gear terminology [5]

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A Appendicies

A.1 Transmission Drawings

The below diagrams[23] of the gear are hoped to, together the prose descriptions in the Preliminary design, give context to the transmission design developed, it's stages and the positioning of the system within the helicopter body.

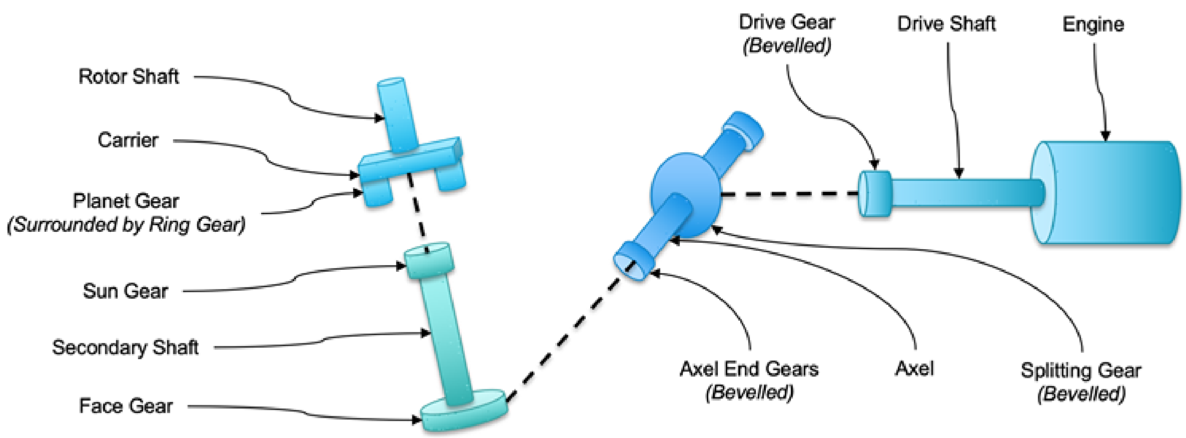


Figure 8: *Transmission Sub-Assembly and Part Diagram. The same colour coding is used for these sub-assemblies throughout this family of images.*

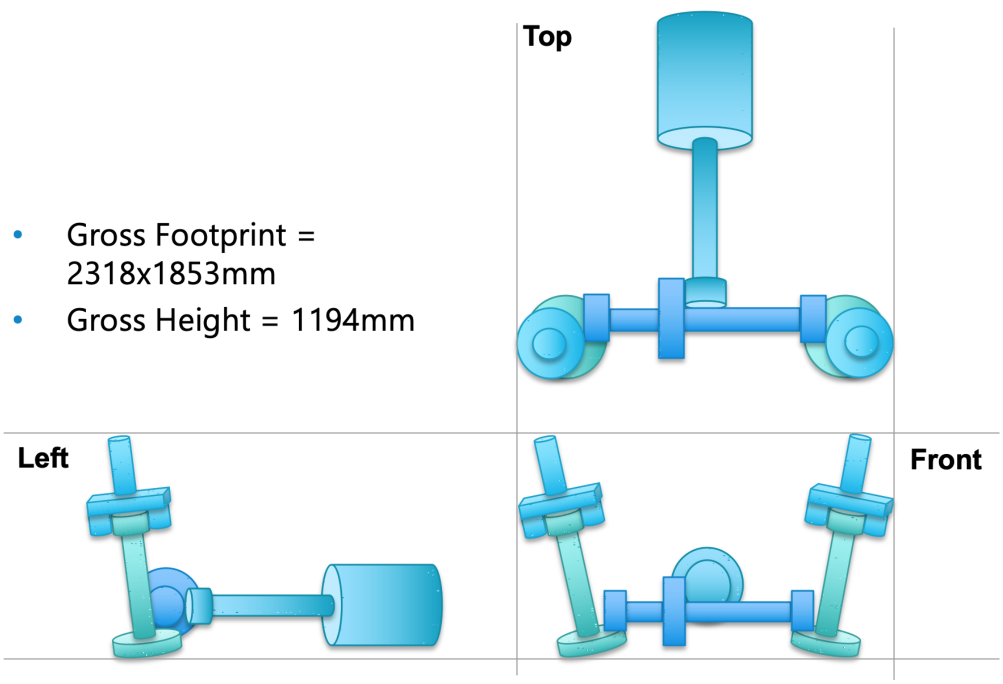


Figure 9: *A Three View Drawing of the Transmission Assembly. Note here that the engine is incorrectly displayed as being in-line, not at a 10° declination.*

[23]These diagrams are admittedly rudimentary, being constructed in Microsofts' *PowerPoint*. Additional more high-spec CAD diagrams can be found elsewhere in this Report.

The below CAD renders[24] show the placement of Auxilary systems around the Transmission, in particular the curves 'S-Duct' geometry of the engine inlet and also the aerodynamic coverings around the pylons to minimise the drag penalty from second main Rotor.

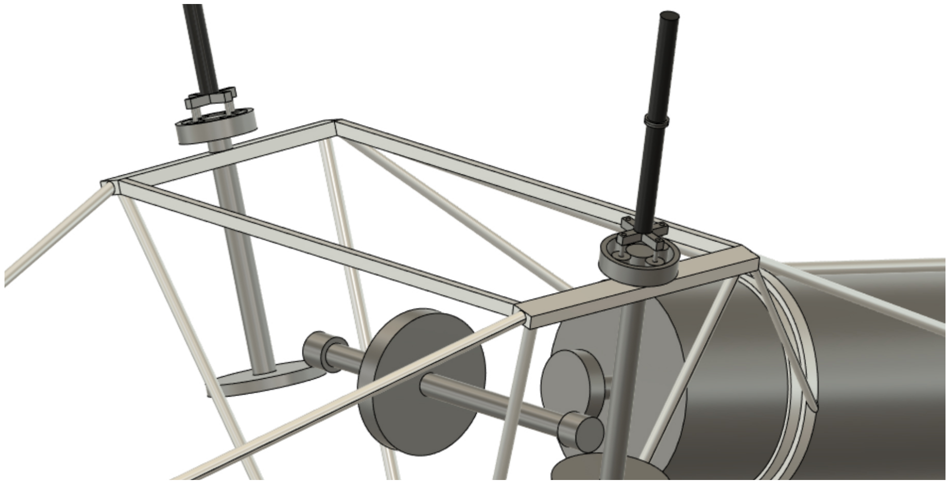


Figure 10: *Transmission, with engine ducting hidden, shown inside the airframe. The planetary gears and supporting bearings on the upper sections are clearly visible. All gears are rendered as running circles only.*

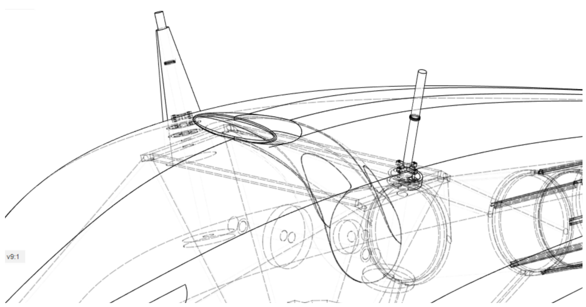


Figure 11: *Hidden Line Drawing, showing the situation of the Pylons and Intake on the Rotorcraft Exterior*

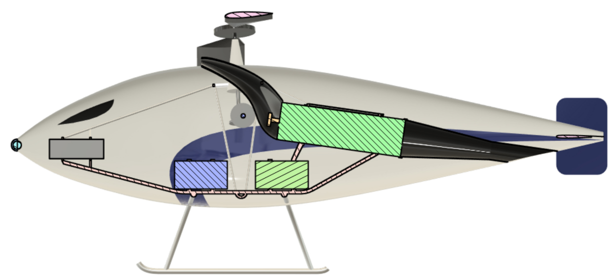
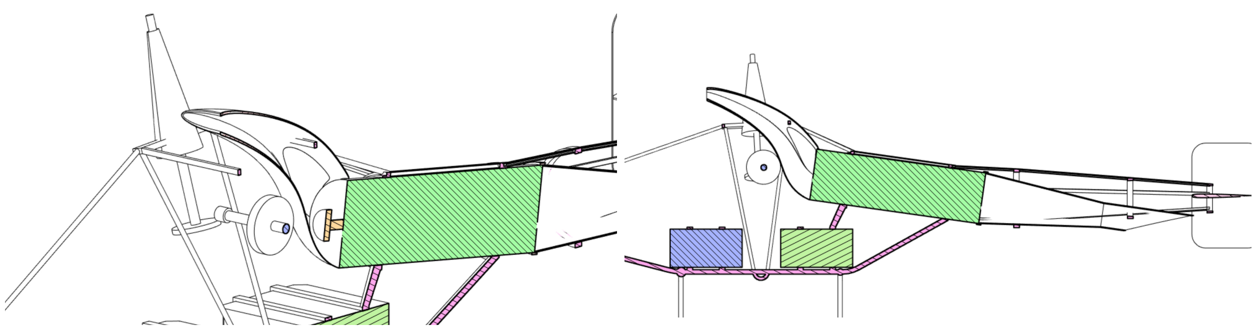


Figure 12: *Side profile of the design, complete with aeroshell covering, and blades.*



(a) *Angled Sectional View of Engine duct near the transmission.*

(b) *Fuel Tank and Engine Sectional Side View*

Figure 13: *Accessories placement & Engine Ducting around the Transmission.*

[24]The Author would like pay due credit to the CAD team [1] for their work in producing some of these rendering remotely in *AutoDesk Fusion 360*, given hand based sketches and relevant data, in co-operation with the author due to the remote working and technological limitations imposed by the 2020 UK Coronavirus Lockdown.

A.2 Gear Nomenclature

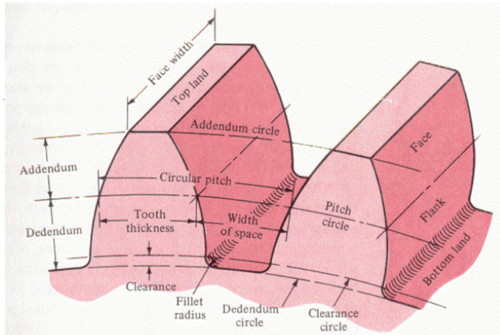


Figure 14: *Nomenclature used in Gears [3]*

Gears

Gears are an oft underlooked and under appreciated area of Aeronautical Turbo-Machinery, falling in the somewhat murky grey area between Aerospace and Mechanical Engineering. Gears are mechanical devices that take a rotational input and manipulated it, this usually results in a scaling up or down of the rotational speed and can be combined with a change in direction also. The principle of gears essentially trades RPM for torque. Note that it was for this reason that *SynchroCaptures'* clutch was mounted before the first reduction at the stage of highest speed but lowest torque, designing to cope with the considerable RPM (speed) is generally less mass intensive than designing for a large torque (moment or force). In gear terminology there are almost always multiple links in a set, which transfers power. In a reduction set the latter of these will have to be larger to revolve at a lower RPM, this driven gear is referred to as the *pinion* and driving gear retaining the name *gear*.

The ratio of the change in rotational speed between gears is called the *Reduction Ratio* and is usually defined by the number of times the gear or pinion turns with respect to a unit cycle of the other. As we are only focussing on the de-scalling of RPM we will set the pinions value in this ratio to unity. Thus a reduction ratio of 2:1 take two rotations of the drive gear to produce a one turn input. This is usually performed by having the pinion be of a larger diameter - whilst keeping the same tooth spacing for the two devices to mesh. The reduction ratio, m_G can be obtained from taking dividing the numbers of teeth of the gears,N,by dividing their *pitch diameters*, D or their rotational speeds, N - these are constant running circles that are used to model the gears without the complexity of modelling teeth! Take the case of an input gear **1** spinning a larger driven pinion **2**. If the larger gear is twice the diameter, with twice the number of teeth it spins at half the speed for a reduction ratio of 2:1. A key factor is that the gears must mesh - having common teeth spacing and velocity at the point of contact, where mechanical energy is transferred. This is best done by matching the pitch-line-velocity of the gears ($^{\pi dn}/_{12}$).

$$m_G = \frac{N_2}{N_1} = \frac{d_2}{d_1} = \frac{n_1}{n_2} \tag{4}$$

Keyways

The gears in a transmission are generally linked by shafts (Section 3.1). These shafts have to be able to connect to the gears and maintain the constant angular velocity of the common gear and shaft assembly. A keyway is a small inserted that slots in between the shaft and core of the gear in a small cutout that prevents rotational mismatch and unifies the component.

A.2.1 Types of Gear

As mentioned in the body of the report a variety of gears are used, these include bevel gears, face gears and planetary gears, which will be briefly discussed here. Asides from the type of gear there are further details about the gears used that have particular importance and impact on the design.

Bevel Gears

A bevel year is a cone like gear in which the toothed gear area forms the cones sloped surface (i.e.

there exists a draft angle), this surface usually mates with another similar bevelled or straight gear to transfer power from one axis of rotation to another - provided that these axis do meet in a fixed point in space. Bevel gears can also transfer power to skew axis but the process is more complicated. Bevel gears for the mainstay of the transmission presented, with intermediate draft angles designed to increase the **contact ratio** - the amount of the gear/number of teeth meshing at one time instance. If two bevel gears meet to transfer power at 90° - usually with two 45° drafts then the arrangement is called a Miter gear. This arrangement would be used in the drive section of the transmisssion however due to the large incoming RPM a system that is partway between two bevel gears and a bevel and a face gear with a 30°/60° draft angle is used for efficient power transmission (See Figure 25).

Face Gears

A face gear is the term for any gear where the teeth are not located on the external revolving surface, but rather on the gears 'face' - the plane perpendicular to the axis of rotation. These gears are most often driven by a connecting regular gear when an angular power transmission is required, such as from the main drive axle to the rotor shaft. The large pick-up area on the face of the gear allows for better mating and connection than using two shafts with end-gears in the same manner. Two counter rotating face gears either side of a bevelled input shaft is an easy way of obtaining a Co-Axial like drive output. As mechanical transmissions advance it has been seen that face gears are beginning to overtake planetary gear arrangements in ebing the primary reduction device for high-reduction ratio helicopter transmissions due to the lower number of moving parts and simpler overall design [24].

Planetary Gears

A planetary or eppicyclic gearset are perhaps the most misunderstood of all gears. A planetary gear set can do many things, as well as descaling torque or RPM such as invert the direction of rotation. A planetary arrangement consists of three parts (See Figure 15): In our case a central driven *Sun* gear that is orbited by several *Planet* gears held within a fixed *Ring* gear. Depending on the motional state of this ring different power configurations can be obtained. With the ring fixed the output is scaled down and in the same direction as the input - this arrangement is very common in helicopters with most marker transmissions using a planetary gearset below their main rotors due to the very large reduction rations and mechanical support they can afford.

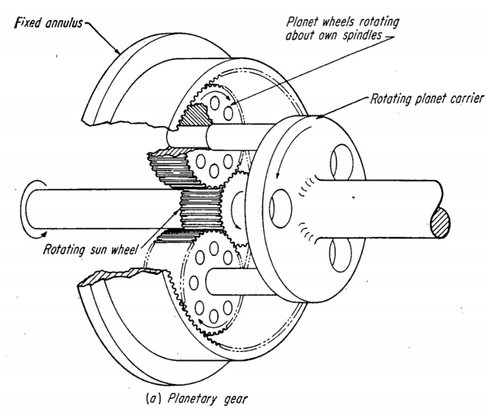


Figure 15: *Fixed Annulus (ring) Planetary Gear simmilar to that used in this transmission [4]*

A.2.2 Further Gear Features

Spur Gears

A spur gear is the terminology for a *classic* gear with straight cut radial teeth - similar to those in children's toys. The simplest of all gears spur gears are notoriously noisy and ineffecient at transmitting mechanical loads.

Helical/Spiral Gears

Helical hears are similar to spur gears except for having curved teeth (a *helix* angle) which are swept along the outer surface of the gear (See Figure 17). This feature greatly reduces the noise produced by the gear and increases contact ratio and efficiency at the price of producing a small axial force along the axis of revolution of the gear, this thrust is usually small for most gears and can be eliminated by the right thrust bearing supports on the connected shafts. Note that the greater contact area for spiral gears means that overall smaller 'coarser' gears can be used (coarseness refering to the diametric pitch ebing smaller) which reduces transmission weight.

A.2.3 Failure Mechanisms of Gears

"Aerospace gearing is sized on three considerations: bending fatigue, surface compression (Hertz stress), and scuffing (scoring) resistance." [24]

Figure 16: *Common gear failure zones [5]*

Figure 17: *Helical Pinion thatas failed from an overload [4]*

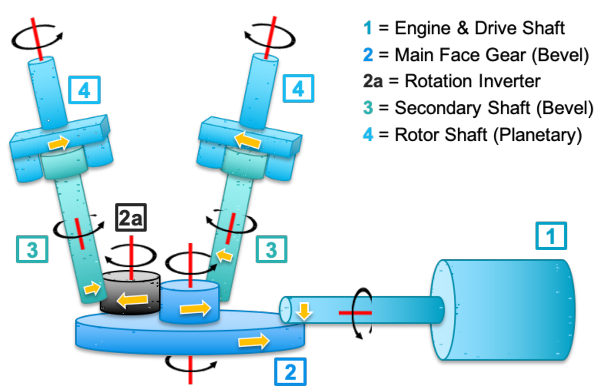
There a variety of unique and fascinating failure methodologies for gears [4], as per any structural design the optimal point for the lightest concept occurs when the failure stresses are all almost equal. For a gear this is when the bending stress and compressive stresses are relatively equal[25]. Of these two major failure types bending is the most severe as it leads to tooth failure which can result is sever internal damage from debris.

An additional failure methodology to be aware of and that should be inspected for is scuffing (scoring), which occurs when the moving gears are under too high of a contact load and scratch each others surface, occasionally welding themselves together. To prevent this in *SynchroCapture* a high temperature running lubricant is pumped between the gears and the detailed design would be such that the contact pressure is never high enough to initial scoring in normal operating conditions.

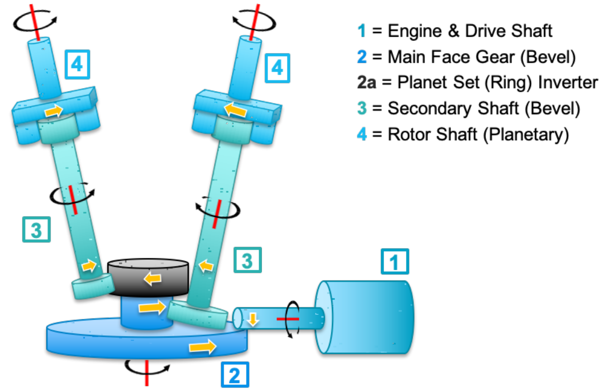
A.3 Design Iteration

For the sake of brevity the final gearbox design selected and only some of the most important reasoning behind this decision was mentioned in the body of this report (Section 3.1). For the purpose of further justification and to explain a little more of the design process this Appendix Section will list out other design considerations configured and why they were finally eliminated as feasible possibilities. Two precursor designs can be seen in Figure 18. These designs were postulated to solve the issue of converting a single engine output at 12000 RPM to two outputs at the rotor speed of around 500 RPM, spinning in the opposite direction - such that the right rotor spins clockwise when viewed from above [28].

[25] A through and comprehensive analysis of different failure mechanisms is given in Dudley's Gear Handbook [4]. If the reader is further curious then Dudley's is advised as the port of call for all further gear related enquiries



(a) *Additional (Asymmetric) inverter-gear concept*



(b) *Counter-rotating planetary gear proposal*

Figure 18: *Rejected designs for the synchropter transmission, based on a large single main rotor-esque face gear.*

The **inverter gear** concept can be seen in Figure 18a, an interesting workaround this design was mechanically simple but was however asymmetric, having the 2a on one side, this would cause a shift in the CG away from the centre-line. A **central planetary gearbox** was also evaluated (Figure 18b) however the need to extract the opposing rotation from a non-fixed ring epicyclic gearbox also brought about asymmetries in addition to the increased mechanical losses, a lack of redundancy and no potential for separate running and realisation that in this arrangement the two outputs may very well be at different speeds.

The elegant workaround solution can be seen in Figure 19 Solution – the use of a common axle. This design is explored in Section 3.1 and notably allows for one rotor still to run in event of Shaft 3 or 4 failing as well as featuring a “better” system for providing the correct gear reduction.

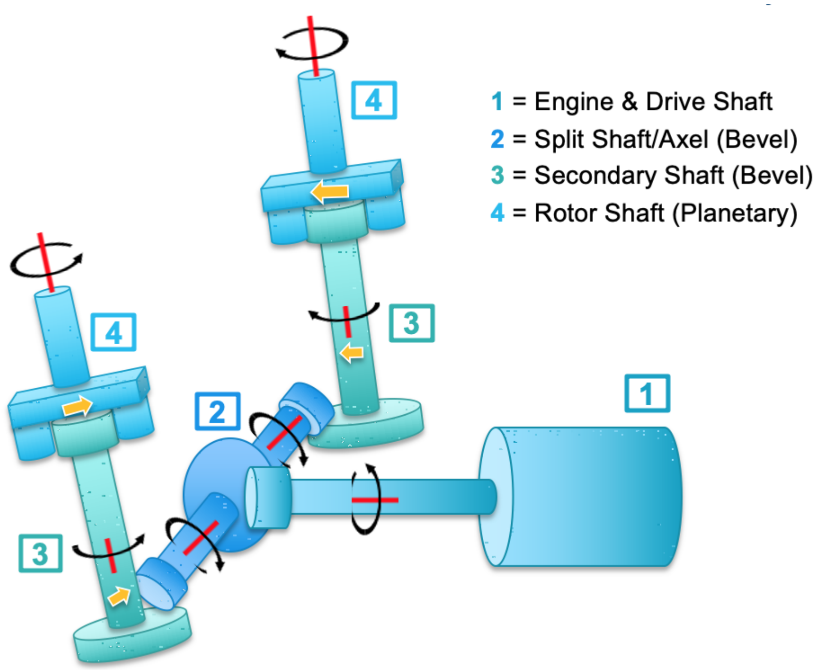


Figure 19: *Final Gearbox Layout, showing rotational directions*

A.4 Design Space: Architecture of the Synchropter Transmission

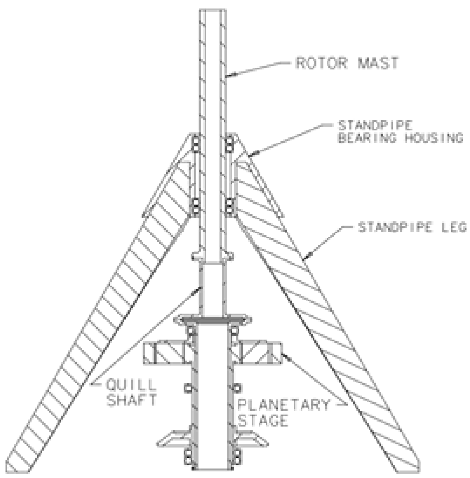


Figure 20: *Standpipe structure from a comparable Synchropter Design from the University of Maryland[6]*

Although there are **several aerodynamic deficits** to the synchropter design when compared to conventional Single Main Rotor (SMR) design; such as a higher induced flow, since the rotors work in each other's downwash and a lower hover efficiency because part of the thrust is lost in the angling of the blades and interactions; there are **tangible benefits** in terms of the aircrafts footprint as this layout allows for a **very low disc loading**. Despite the offset in the rotors (which leads to some power loss) the more compact nature of a synchropter layout lent itself better to the role a UAV at sea. In particular two angled rotors are beneficial if one wants to land on a ship rolling on the waves as significant stabilising force can be transferred to the skids. However, in analysis it has been found that this configuration can exhibit worse off cruise performance, compared it a single main rotor, as having two hubs and rotor stems adds higher parasitic drag.

From the point of view of the propulsion and transmission the concept of having intermeshing bladesets had some advantages over the other designs – of main focus here is that the propulsive transmission is far simpler. As there is **no need for an anti-Torque tail** (blade travel is near symmetric) there is no secondary bleed of torque from the engine through a tail boom to a tail rotor as would be the case in a single main rotor design and further **no need for the complex counter-rotating shafts of a co-axial design**, where the two spinning drive shafts for the rotors are inside each other. Further given the operating scenario and environment (the sea) - and to lower cost related articles that the reduced mechanical complexity of having only one gearbox that feeds two main rotors brings this design was decided on. The reduction in mechanical complexity can be seen in the layout diagrams in Figure 8.

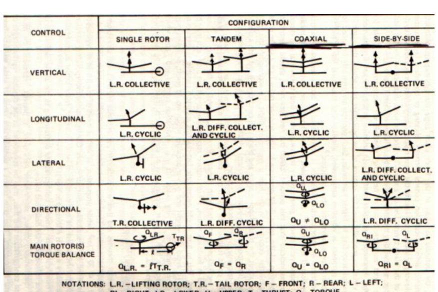


Figure 21: *Control schemes for different rotor layouts [7]*

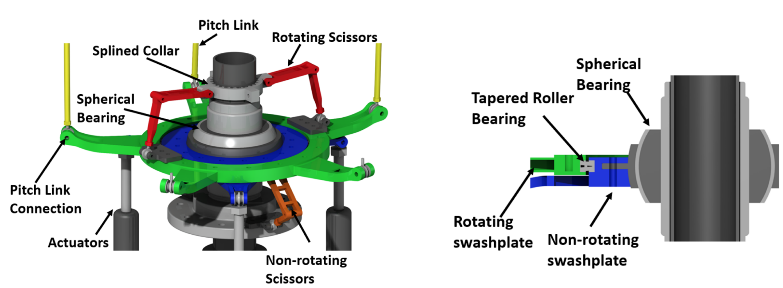
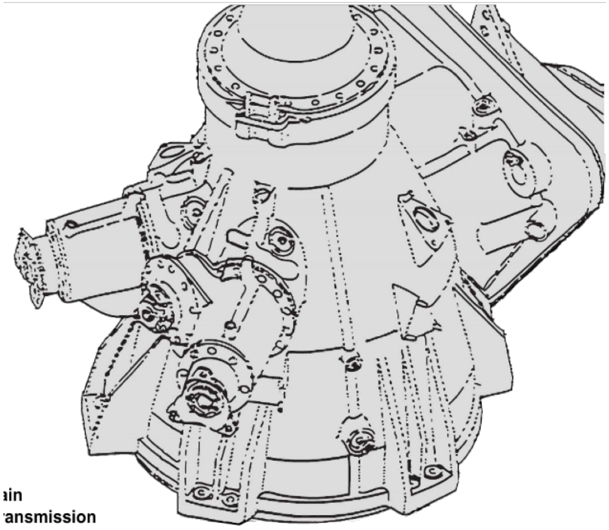
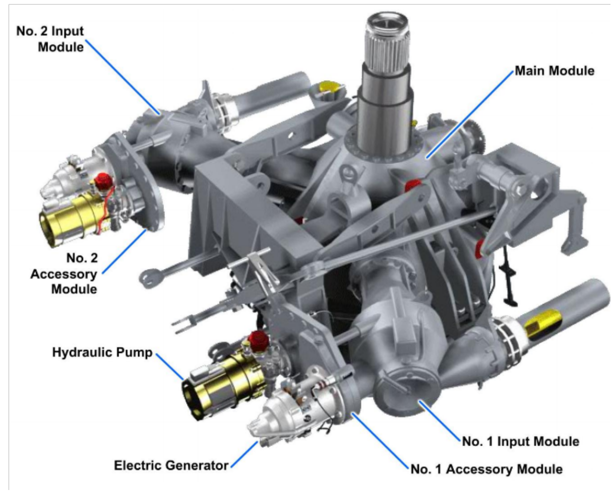


Figure 22: *Swashplate Assembly and Sectional View [8]*

Of note in the design was the angling of the engine, this served as a compact Centre of Gravity (CG) correction. Best practice in rotorcraft stipulates that the CG should be near to the rotor, if not a little in front. The engine is a significant mass rear of the rotors hence has the capability to shift the CG is slightly aft although it does react some of the fuselage components. The placement of the engine minimsed it's contribution. Likewise the fuel tanks were situated at the bottom of the fuselage, below the gearbox and winch-mechanism to minimise trim shift as fuel mass changes.



(a) *CH-53 Main Gearbox Casing [24]*



(b) *SH-60 Main Gearbox and Accessories power porting [24]*

Figure 23: *Images of the gearbox casing and accessories accommodations of two other sea-based helicopters. Whilst these helicopters are SMR configuration the end results (Sections 5.1 & 3.5) are comparable.*

A.5 Saftey Factors

"The need to save weight means that every possible piece of material needed to contribute it's full quota of strength and stiffness [and performance efficiency] to the design." [37]

In the design process shown in Section 3.1 the following design safety factors and definitions were used when considering maximal loading components needed to withstand. Other nominal load cases and factors are specified where relevant [21]. Note that some more critical parts such as the swashplates, which form a *Single Point of Failure* would in further analysis be expected to perform to additional loading qaulities.

- **Limit Load** - The maximum load that the rotorcraft would be expected to experience in normal operation
- **Ultimate Load** - 1.5 times the limit load. The structure and load bearing components must support the ultimate load before it fails i.e the limit with a safety factor

A.5.1 Design life

Whilst not necessarily a feature of explicit safety factors there are expected lifetimes that can be suggested (either through OEM guidelines or FAR regulations) for components undergoing cyclic loading. As a suggestion the Rotor Hubs are to be rated for at least 5000-7000 operational hours in between full services. In addition gears and tribological components are to be rated for a design life of 11,000 hours[38]. This is 10% more than the standard life for these components and falls inline with FAR 29 guidelines on 'wear and tear'.

A.6 Altitude-Power Lapse Relation

A turboshaft was selected to power *SynchroCapture* due to its higher power-to-weight ratio when compared to an internal combustion engine. Further a turboshaft possesses increased reliability and lower power-lapse when compared a piston engine (which would have to use a complex and heavy compressor and carburation systems to negate this thinning air at altitude), these factors being central within the scope of a high altitude heavy lift design meant that a Turboshaft was far the best option for power delivery. In the course of the conceptual design phase it was necessary to perform an initial engine and thrust sizing, the following derivation for power lapse at altitude was formulated.

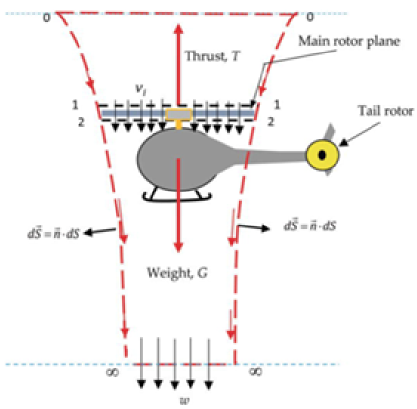


Figure 24: *Control volume for the rotor blade in hover [9]*

Turboshaft & Rotor Scaling :
Through research it was discovered that Helicopters as an aircraft species experience **very** substantial power losses as altitude increases as a result of thinning air for aspiration and resultant decreased lift. (At an altitude of 20,000ft the loss in hover will be shown to be some 0.39 times the horsepower of a rotorcraft operating at sea level [39]). The power output of a Turboshaft decreases linearly with decreasing density, ρ: $P_i/P_{ASL} = \rho_i/\rho_{ASL}$. Here the Power at altitude, P_i is related to the power at sea level, ASL and density ratio. A control volume analysis (Figure 16) of the rotor blade in hover can be conducted to show that the power output of the blade decreases with the square root of density:

$$P = \frac{T^{\frac{3}{2}}}{\sqrt{2\rho A}} \tag{5}$$

Proportional Power to Weight Relationship::
Curve fitting of the relationship between aircraft empty weight W_{Empty}, in kg, and Sling Payload Weight W_{Sling}, also in kg, displayed a linear relationship $W_{\text{Empty}} = 1.34W_{\text{Sling}} + 42.5$. We can obtain a proportionality between the engine power and rotor diameter, D versus the combined weight that needs to be lifted noting that: $\sqrt{A} \propto D$ with a constant density and the thrust equal to the combined weight: $P \times D = (W_{\text{Empty}} + W_{\text{Sling}})^{\frac{3}{2}}$ (for a hover). Further regressional sizing may be used to relate the Power (at sea level) to the new combined weight for a given rotor diameter, in meters. The forecast linear relationship between $P \times D$ and W_{Combined} is $P \times D = 0.0443W_{\text{Combined}}^{\frac{3}{2}} + 2667$ [9]

Conclusion:
From the Control Volume we can see that $P \propto \rho^{-1/2}$ hence its contribution is $P_i/P_{ASL} = \sqrt{\rho_i/\rho_{ASL}}$. The total helicopter output at sea level to produce enough lift from an engine and blade perspective is therefore:

$$P_i = P_{ASL} \cdot \left(\frac{\rho_i}{\rho_{ASL}}\right)^{\frac{3}{2}} \tag{6}$$

For example at 20000ft the density ratio is 0.53, thus when raised to the power three halves this is 0.39. The power required at sea level to achieve a stipulated power, P_i is thus $1/0.39 = 2.56$ times the stipulated power power.

A.7 Design Sketches

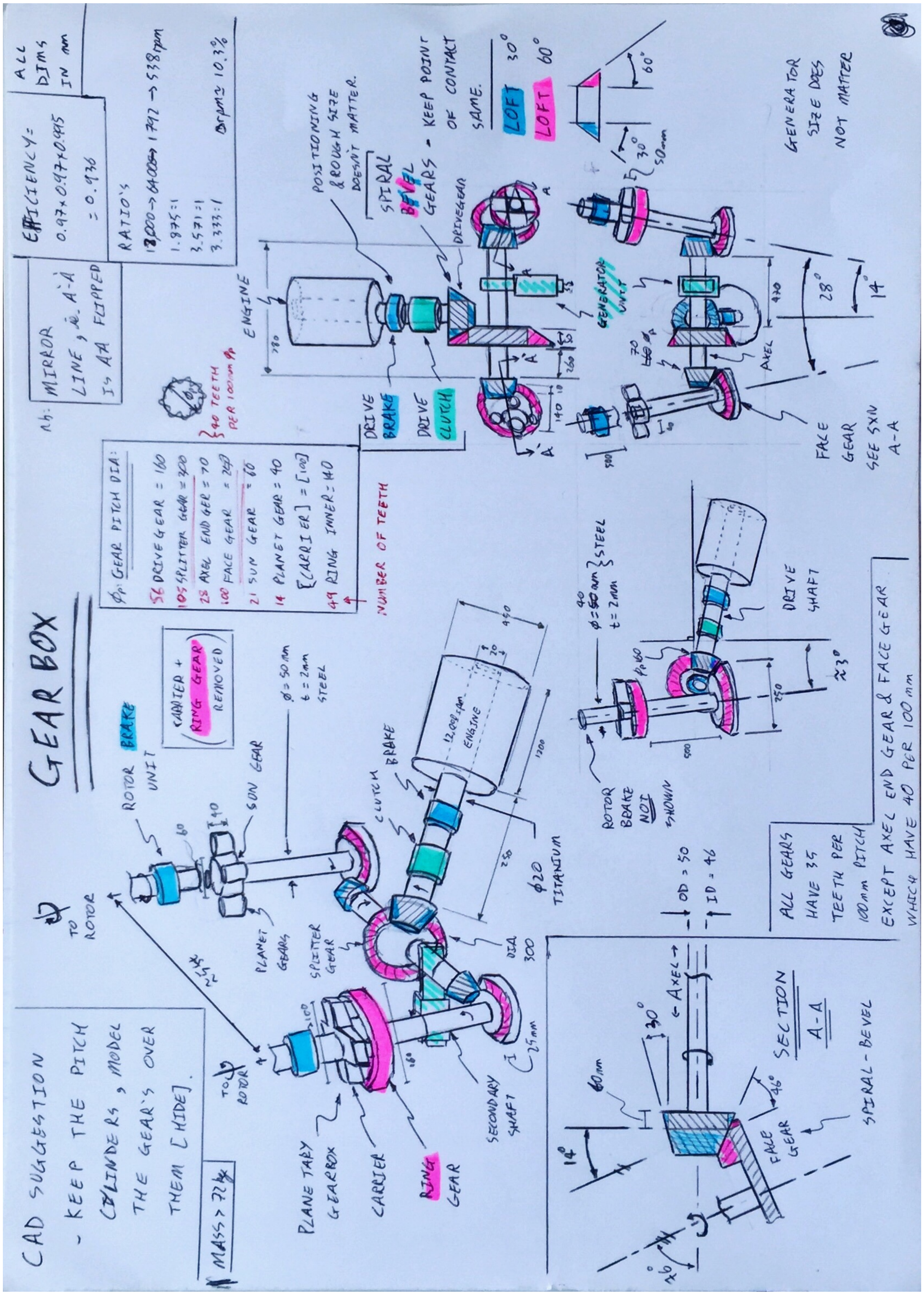


Figure 25: *Sketch-page of the final transmission, highlighting key gear faces and components. A sectional view of the bevel/facegear interface is given.*

B Supporting MATLAB Code

The below script was developed for the purpose of evaluating the designed transmission. This includes a preliminary weight estimation, calculation of reduction ratios, losses in the system and also evaluating the shaft torque and sizing as well as featuring some geometry related to the swashplates and a diagrammatic sketcher and outlines a mass estimation methodology. A script for the plotting of the Mission Profile (Figure 1) was also constructed but is not featured here.

```

% AERO96005 GROUP DESIGN PROJECT
% HELICOPTER_TRANSMISSION.MLX
% - - - - - - - - - - - - - - - - - - - - - - - - - - -
% DETERMINES REQUIRED GEAR RATIOS, EVALUATES TORQUE &
% TORQUE TUBE SIZING AND SEVERAL OTHER TRANSMISSION
% OBJECTS WHEN GIVEN THE REQUIRED ENGINE INPUT & ROTOR
% OUTPUT DETAILS.
% T CROSS 25/05/2020
% - - - - - - - - - - - - - - - - - - - - - - - - - - -

% WORKSPACE BOOK-KEEPING

clear
clc

format longG

% conversions
lbs2kg = 0.453592;
kg2lbs = 2.20462;
mm2ft = 0.00328084;
hp2watt = 745.7;
ftlbs2Nm = 1.36;

% This is purely here to guide the user through the stages in the
% transmission!
% Guides:

% PARTS AND SEQUENCES
% Engine Inlet
% Drive Shaft
% Drive Gear

% Splitting Gear
% Axel Left & Right
% Axel End Gears

% Face Gear
% Secondary Shaft
% Sun Gear

% Planet Gear
% Ring Gear

% Carrier
% Rotor Shaft

%% Inputs: Engine Power
```

```

%INPUT
Factor = 1.5; %Limit Load Factor for ratings
Number_of_Engines = 1;
MTOW = kg2lbs*2907, disp("lbs") %lbs

RPM_Engine = 12000, disp("rpm")     %rpm - 2000 to 8000
Engine_Output_Power = 1600, disp("shp")      %hp
% Torque
T_Engine = 5252*ftlbs2Nm*Engine_Output_Power/RPM_Engine;
T_Engine_safe = Factor*T_Engine, disp('Nm')

%Omar has a 50hp loss due to gears

Drive_Rating = Factor*Number_of_Engines*Engine_Output_Power, disp("shp")


%% Output: Rotor Details

%INPUT
%From design_parameter.xslx
Rotational_frequency=8.1252;    %rev/sec 525rpm
Tip_Speed = 221.2;  %(m/s) - hover
Rotor_Diameter = 26.4; %ft
Disc_Loading = 6; %lb/ft^2

%To English
One_Rotor_Area = 0.25*pi*Rotor_Diameter^2; %ft^2 effective disk area
Tip_Speed = Tip_Speed*3.28084; %ft/s

%Basic Info
Desired_Rotor_Revs = Rotational_frequency*60, disp("rpm")      %rpm
Revolution_Reductuion_Ratio = Desired_Rotor_Revs/RPM_Engine;

%PRINTOUT
Initial_Figure = 1/Revolution_Reductuion_Ratio;
Ratio = ['The gearbox scales for ', num2str(Initial_Figure), ':1'];
disp(Ratio)

%Each rotor gear to be rated to take half the engine power
%Rotor_Separation = 2; %m


%% Design Space

%Design Space
HP_reqMR = 225*0.514*MTOW*9.81*0.00134102; %hp, Main Rotor Power Required - ...
    conveeted (power = force x velocity)

HP_engMRP = Engine_Output_Power; %hp, Engine Maximum Rated Power (???)
HP_reqAccess = 60; %hp, Accessory Power Required  - 60 is low 120 is a baseline

Th_MR = MTOW; %lb, Main Rotor Thrust
n_MR = Desired_Rotor_Revs;%rpm, Main Rotor Speed
n_eng = RPM_Engine; %rpm, Engine Speed

%% Weigth Estimates

%Zuzanna and Wayne's Estimate
```

```
W_Estimate = lbs2kg*577, disp("kgs")

%NASA Technical Report 82-A-10, A Comparative Study of Soviet vs. Western
%Helicopters Part 2.17
a_mr = 1 %adjsutment factor
z_mr = 2 %number of stages in main rotor drive
k_t = 1.2 %configuration factor, 1 for the main rotor (unknown what ours is)

W_DriveSystem_MainRotor = 250*a_mr*( (HP_engMRP/n_MR) * z_mr^0.25 * k_t)^0.67, ...
    disp("lbs")
W_DriveSystem_MainRotor_Metric = lbs2kg*W_DriveSystem_MainRotor, disp("kgs")

num_gearboxes = 3;
L_DriveShaft_Horz = 5;  %The horizontal distance between rotor hubs (feet)
num_DriveShafts = 2; %excludes rotor shaft

W_gb = 172.7*((HP_engMRP/n_MR)^0.7693)*num_gearboxes^0.1406;
W_shafting = 1.152*(HP_reqMR/n_MR)*L_DriveShaft_Horz^0.8829*num_DriveShafts;

W_DriveSystem_MainRotor_2 = W_shafting + W_gb, disp("lbs")
W_DriveSystem_MainRotor_Metric_2 = lbs2kg*W_DriveSystem_MainRotor_2, disp("kgs")

%% Sizing (CAD in MATLAB)
%all lengths in this section in mm
%all angles in degrees

% ROTOR OFFSET
Angle_Rotor_OffsetVert = 14, disp("degrees")

rho = 0.000008 %kg/mm3 steel

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% ENGINE PARTS
% For gear parts the diameters given CONSTANT ROLLING PITCH DIAMETERS

% Engine Inlet - for completeness
L_Engine = 1200;
Dia_Engine = 450;
Engine_declination_Angle = 10; %degrees
%these are not used in calcs

% Engine RPM
RPM_Engine

% Drive Shaft
L_DriveShaft = 250;
Dia_DriveShaft_Outer = 20;
Thickness_DiveShaft = 0; %is solid
Mass_est_DriveShaft = rho*(0.25*pi*(Dia_DriveShaft_Outer^2))*L_DriveShaft;

% Drive Gear
Depth_DriveGear = 40;
Dia_DriveGear = 160;
Mass_est_DriveGear = rho*(0.25*pi*(Dia_DriveGear^2))*Depth_DriveGear;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% AXEL PARTS

% Splitting Gear
```

```
Depth_SplittingGear = 50;
Dia_SplittingGear = 300;
Mass_est_SplittingGear = rho*(0.25*pi*(Dia_SplittingGear^2))*Depth_SplittingGear;

%GEAR REDUCTION
Ratio1 = Dia_SplittingGear/Dia_DriveGear;
Ratio_1 = ['The FIRST (AXEL) link scales for ', num2str(Ratio1), ':1'];
disp(Ratio_1)
RPM_Axel = (1/Ratio1)*RPM_Engine
%Stage efficiency
Eff_1 = 0.97 %plus as based on double reduction bevel

% Axel Left & Right
L_Axel_Left = 470;
L_Axel_Right = L_Axel_Left-Dia_DriveGear-Depth_SplittingGear;
L_LeftandRight =L_Axel_Left+L_Axel_Right+Depth_SplittingGear
Dia_Axel_Outer = 50;
Thickness_Axel = 2;
Dia_Axel_Inner = Dia_Axel_Outer - 2*Thickness_Axel;
Mass_est_Axel_LandR = ...
    rho*(0.25*pi*(Dia_Axel_Outer^2-Dia_Axel_Inner^2))*(L_Axel_Left+L_Axel_Right);

% Axel End Gears
Depth_AxelEndGear = 60;
Dia_AxelEndGear = 70;
Mass_est_AxelEndGear = rho*(0.25*pi*(Dia_AxelEndGear^2))*Depth_AxelEndGear;

Angle_Axel2Secondary = 90+Angle_Rotor_OffsetVert;
Angle_ForwardCant = 3;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% SECONDARY SHAFT PARTS

%Face Gear
Depth_FaceGear = 25;
Dia_FaceGear = 250;
Mass_est_FaceGear = rho*(0.25*pi*(Dia_FaceGear^2))*Depth_FaceGear;

%GEAR REDUCTION
Ratio2 = Dia_FaceGear/Dia_AxelEndGear;
Ratio_2 = ['The SECOND (BEVEL-FACE) link scales for ', num2str(Ratio2), ':1'];
disp(Ratio_2)
RPM_Secondary = (1/Ratio2)*RPM_Axel
%Stage efficiency
Eff_2 = 0.97 %plus as based on double reduction bevel

% Secondary Shaft
L_SecondaryShaft = 500;
Dia_SecondaryShaft_Outer = 50;
Thickness_SecondaryShaft = 2;
Dia_SecondaryShaft_Inner = Dia_SecondaryShaft_Outer - 2*Thickness_SecondaryShaft;
Mass_est_SecondaryShaft = rho*(0.25*pi*(Dia_SecondaryShaft_Outer^2- ...
    Dia_SecondaryShaft_Inner^2))*L_SecondaryShaft;

% Sun Gear
Depth_SunGear = 40;
Dia_SunGear = 60;
Mass_est_SunGear = rho*(0.25*pi*(Dia_SunGear^2))*Depth_SunGear;
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% ROTOR SHAFT PARTS

%Planet Year
num_planets = 4
Depth_PlanetGear = Depth_SunGear;
Dia_PlanetGear =40;
Mass_est_PlanetGears = ...
    rho*num_planets*(0.25*pi*(Dia_PlanetGear^2))*Depth_PlanetGear; %all of them ... in one

% Carrier
L_Carrier = 50;
Dai_Carrier = Dia_SunGear+(0.5*(num_planets/2)*Dia_PlanetGear); %for a 4 planet ... system
Mass_est_Carrier = rho*(0.25*pi*(Dai_Carrier^2))*L_Carrier;

%Ring Gear
Dia_RingGear_Inner = Dia_SunGear+2*Dia_PlanetGear;
Depth_RingGear = Depth_SunGear;
Thickness_RingGear = 10;
Dia_RingGear_Outer = Dia_RingGear_Inner + 2*Thickness_RingGear;
Mass_est_RingGear = ...
    rho*(0.25*pi*(Dia_RingGear_Outer^2-Dia_RingGear_Inner^2))*Depth_RingGear;

%GEAR REDUCTION
Ratio3 = 1+Dia_RingGear_Inner/Dia_SunGear;
Ratio_3 = ['The THIRD (PLANETERY) link scales for ', num2str(Ratio3), ':1'];
disp(Ratio_3)
RPM_Rotor = (1/Ratio3)*RPM_Secondary
%Stage efficiency
Eff_3 = 0.995 %plus as based on 5:1 PLANETERY

%Rotor Shaft
L_RotorShaft = 500;
Dia_RotorShaft_Outer = 40;
Thickness_RotorShaft = 2;
Dia_RotorShaft_Inner = Dia_RotorShaft_Outer - 2*Thickness_RotorShaft;
Mass_est_RotorShaft = ...
    rho*(0.25*pi*(Dia_RotorShaft_Outer^2-Dia_RotorShaft_Inner^2))*L_RotorShaft;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% OVERVIEW

%Compare to original
Desired_Rotor_Revs
Δ_RPM=RPM_Rotor-Desired_Rotor_Revs
disp("Positive indicates we can spin faster than we want, to account for losses")
Percentage_OverSpin = 100*((RPM_Rotor/Desired_Rotor_Revs)-1)
%GearBox efficiency - cpompounded from Dudley Gear Handbook
Total_Efficiency = Eff_1*Eff_2*Eff_3
RPM_withefficiency = Total_Efficiency*RPM_Rotor
%Gearbox_Losses = 1-Total_Efficiency

%YIELDED GLOBAL DIMESIONS

L_EngineInput = L_DriveShaft+Depth_DriveGear
L_Baseline = (L_EngineInput*cosd(Engine_declination_Angle))+Dia_AxelEndGear
L_Baseline_with_engine = L_Baseline+(L_Engine*cosd(Engine_declination_Angle))
```

```
Slope_Baseline_with_engine = ...
    L_EngineInput+Dia_AxelEndGear+L_Engine*cosd(Engine_declination_Angle)

L_Axel_ASM = L_Axel_Left+L_Axel_Right+Depth_SplittingGear+(2*Depth_AxelEndGear)
L_SecondaryShaft_ASM = Depth_FaceGear+L_SecondaryShaft+Depth_SunGear
L_RotorShaft_ASM = Depth_PlanetGear+L_Carrier+L_RotorShaft
L_Slope = L_RotorShaft_ASM+L_SecondaryShaft_ASM-Depth_SunGear

Mass_est_EngineInputASM = Mass_est_DriveGear+Mass_est_DriveShaft
Mass_est_AxelASM = ...
    Mass_est_Axel_LandR+Mass_est_SplittingGear+(2*Mass_est_AxelEndGear)
Mass_est_SeondaryShaftASM = ...
    Mass_est_FaceGear+Mass_est_SecondaryShaft+Mass_est_SunGear %for one side
Mass_est_RotorShaftASM = ...
    Mass_est_PlanetGears+Mass_est_RingGear+Mass_est_RotorShaft %for one side

Mass_Transmission_ASM = ...
    Mass_est_EngineInputASM+Mass_est_AxelASM+2*(Mass_est_SeondaryShaftASM+ ...
    Mass_est_RotorShaftASM), disp('kg')

Width_Baseline = L_Axel_ASM+Dia_FaceGear*sind(Angle_Rotor_OffsetVert)
Net_Length_Midpoints = (L_Engine*cosd(Engine_declination_Angle))+Dia_AxelEndGear ...
    +(L_Slope*sind(Angle_ForwardCant)) %Total Length of Asm
Net_Width_Midpoints = L_Axel_ASM+2*(0.25*Dia_FaceGear)+ ...
    2*(L_Slope*sind(Angle_Rotor_OffsetVert)) %Rotor Separation

Rotor_Separation = mm2ft*Net_Width_Midpoints, disp('ft')

Max_Length_Transmission_ASM = Net_Length_Midpoints+2*(0.5*Dia_RingGear_Inner)
Max_Width_Transmission_ASM = Net_Width_Midpoints+2*(0.5*Dia_RingGear_Inner)
Max_Height_Transmission_ASM = ...
    L_Slope*cosd(Angle_Rotor_OffsetVert)+Dia_FaceGear*sind(Angle_Rotor_OffsetVert)


%% RPM Plots

% LENGTH SERIES
L_Driving = L_DriveShaft+Depth_DriveGear; %excludes engine
% operate at
RPM_Engine;

%these next two numbers should be the same lol...
L_Axel_Driven_Left = (0.5*Dia_SplittingGear)+L_Axel_Left;
L_Axel_Driven_Right = (0.5*Dia_SplittingGear)+Depth_SplittingGear+L_Axel_Right;
% operate at
RPM_Axel;

L_SecondaryShaft_Driven = Depth_SunGear+L_SecondaryShaft+(0.5*Dia_FaceGear); ...
    %includes sun gear
% operate at
RPM_Secondary;

L_RotorShaft_noplanet = L_RotorShaft+L_Carrier; %excludes planet gear
% operate at
RPM_Rotor;

TotalLength_Left = ...
    (L_Driving+L_Axel_Driven_Left+L_SecondaryShaft_Driven+L_RotorShaft_noplanet);
Lengths_Left = (0:1:(TotalLength_Left-1));
```

```

TotalLength_Right = ...
    (L_Driving+L_Axel_Driven_Right+L_SecondaryShaft_Driven+L_RotorShaft_noplanet);
Lengths_Right = (0:1:(TotalLength_Right-1));

%Aim
Aim = Desired_Rotor_Revs*ones(1,length(Lengths_Left));

% RPM SERIES
RPM_Left = [RPM_Engine*ones(1,L_Driving), RPM_Axel*ones(1,L_Axel_Driven_Left),...
    RPM_Secondary*ones(1,L_SecondaryShaft_Driven),RPM_Rotor* ...
        ones(1,L_RotorShaft_noplanet)];
RPM_Right = [RPM_Engine*ones(1,L_Driving), RPM_Axel*ones(1,L_Axel_Driven_Right),...
    RPM_Secondary*ones(1,L_SecondaryShaft_Driven),RPM_Rotor* ...
        ones(1,L_RotorShaft_noplanet)];
tsize = 12
figure
plot(Lengths_Left,RPM_Left,"linewidth",2,"color",[0 0.4470 0.7410])
hold on
plot(Lengths_Right,RPM_Right,"linewidth",2,"linestyle",':',"color",[0.8500 ...
    0.3250 0.0980])
hold on
plot(Lengths_Left,Aim,"linewidth",0.5,"color",'k')
hold on

title({'Operational RPMs of Various Transmission Segments','from Engine Outlet ...
    to Rotor Hub'},'FontSize',tsize)
ylabel("Rotational Speed, n / RPM",'FontSize',tsize)
xlabel("Distance along Transmission Path, l / mm",'FontSize',tsize)
ylim([0 14000])
xlim([0 2200])
grid on

legend('Left Hand Load Path','Right Hand Load ...
    Path','Location',"northeast",'FontSize',tsize)

text(0.02*TotalLength_Left, 1.9*Desired_Rotor_Revs,'Aim RPM','FontSize',tsize)
text(0.02*TotalLength_Left, 1.04*RPM_Engine,'Drive Shaft','FontSize',tsize)
text(0.28*TotalLength_Left, 0.57*RPM_Engine,'Axel','FontSize',tsize)
text(0.50*TotalLength_Left, 0.2*RPM_Engine,'Secondary Shaft','FontSize',tsize)
text(0.82*TotalLength_Left, 1.9*Desired_Rotor_Revs,'Rotor Shaft','FontSize',tsize)

hold off

%% TEETH NUMBERS

TeethPer100mil=35;
TeethPer100mil2=40;

NumTeeth_DriveGear = (Dia_DriveGear/100)*TeethPer100mil
NumTeeth_SplittingGear = (Dia_SplittingGear/100)*TeethPer100mil
NumTeeth_AxelEndGear = (Dia_AxelEndGear/100)*TeethPer100mil2
NumTeeth_FaceGear = (Dia_FaceGear/100)*TeethPer100mil2
NumTeeth_SunGear = (Dia_SunGear/100)*TeethPer100mil
NumTeeth_PlanetGear = (Dia_PlanetGear/100)*TeethPer100mil
NumTeeth_RingGear = (Dia_RingGear_Inner/100)*TeethPer100mil

%% Transmission Calculations

```

```
% CALCULATION FOR MIN THICKNESS

% Solid bar
% Shear stress max
tau_steel = 250000000; %200  300   MPa
tau_ti = 760000000; %760 MPa

D_req_outputshaft = 2*1000*((T_Engine_safe/(tau_ti*(pi/2)))^(1/3)), disp('mm')

% Hollow bar
D_Outer_hollow = 50 %mm
D_Inner = 2*1000*((D_Outer_hollow/(2*1000))^4 - ...
    (T_Engine_safe*(D_Outer_hollow/(2*1000))) / (tau_steel*(pi/2)))^(1/4);
Thk_req_hollow = 0.5*(D_Outer_hollow-D_Inner)

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% Swashplate sizing and calculations

% This code works out various parameters for the swashplates and hodefully
% plots them too

% MAST
h_Mast = 100; %set an arbitrary Height of the mast we want to plot bolow the axes
r_Mast = 100;

% ROTORSHAFT
h_RotorShaft = 160;
L_RotorShaft; %500

Dia_RotorShaft_Outer %40mm
Rad_RotorShaft = 0.5*Dia_RotorShaft_Outer;
Num_Actuators = 4;
Clearance = 10;

% ACTUATORS
Actuator_Pivot = 75

h_Actuator = 50 ;
h_Actuator_L = h_Actuator ;
h_Actuator_R = h_Actuator ;
w_Actuator = 20 ;
r_Actuator_Inner = Actuator_Pivot - (0.5*w_Actuator);

ext_Actuator_L = 0 ;
ext_Actuator_R = 0 ;

% STATIONARY PLATE
h_StationaryPlate = 25 ;
r_StationaryPlate_Outer = 85;
r_StationaryPlate_Inner = Rad_RotorShaft+Clearance;
w_StationaryPlate = r_StationaryPlate_Outer-r_StationaryPlate_Inner

% ROTARY PLATE

h_RotaryPlate = h_StationaryPlate ;
r_RotaryPlate_Outer = r_StationaryPlate_Outer;
r_RotaryPlate_Inner = Rad_RotorShaft+Clearance;
w_RotaryPlate = r_RotaryPlate_Outer-r_RotaryPlate_Inner;
```

```

% BEARING

h_Bearing = 20;
w_Bearing = h_Bearing;
r_Bearing_Inner = r_RotaryPlate_Inner

h_Swashplate_ASM = h_Actuator + h_StationaryPlate + h_RotaryPlate;

% PITCH LINK
Connection_Pitchlink_Swashplate_x = r_RotaryPlate_Outer-(0.5*h_RotaryPlate);
Connection_Pitchlink_Blade_x = 80;
Connection_Pitchlink_Swashplate_y = h_Swashplate_ASM-(0.5*h_RotaryPlate);
Connection_Pitchlink_Blade_y = 180;

Pitchlink_x= [Connection_Pitchlink_Swashplate_x, Connection_Pitchlink_Blade_x]
Pitchlink_y= [Connection_Pitchlink_Swashplate_y, Connection_Pitchlink_Blade_y]

%Rotor
w_Hub = 100;
h_Hub = 40

Blade_x_Bot=[w_Hub/2 , 3*w_Hub]
Blade_y_Bot=[h_RotorShaft , h_RotorShaft+0.5*h_Hub]
Blade_x_Top=[w_Hub/2 , 3*w_Hub]
Blade_y_Top=[h_RotorShaft+h_Hub , h_RotorShaft+0.7*h_Hub]

%Plotting - for the fun of it

Axis_x = [0,0];
Axis_y = [-h_Mast,h_RotorShaft+150];

figure

% mast
Mast =rectangle('Position',[-r_Mast -h_Mast 2*r_Mast ...
    h_Mast],"linewidth",2,'EdgeColor',[0 0.4470 0.7410])
hold on
% rotorshaft
RotorShaft =rectangle('Position',[-Rad_RotorShaft -h_Mast Dia_RotorShaft_Outer ...
    h_Mast+h_RotorShaft],"linewidth",2,'EdgeColor',[0.8500 0.3250 0.0980], ...
    "LineStyle","-")
hold on
% actuator R
A_R =rectangle('Position',[r_Actuator_Inner 0 w_Actuator ...
    h_Actuator_R],'Curvature',0.4)
hold on
A_R.FaceColor = [0.4660 0.6740 0.1880];
A_R.EdgeColor = [0 0.4470 0.7410];
A_R.LineWidth = 2;
% actuator L
A_L =rectangle('Position',[-(r_Actuator_Inner+w_Actuator) 0 w_Actuator ...
    h_Actuator_L],'Curvature',0.4)
hold on
A_L.FaceColor = [0.4660 0.6740 0.1880];
A_L.EdgeColor = [0 0.4470 0.7410];
A_L.LineWidth = 2;

% stationary plate
```

```
rectangle('Position',[-(r_Actuator_Inner+w_Actuator) h_Actuator_R ...
    2*r_StationaryPlate_Outer h_StationaryPlate],"linewidth",2,'EdgeColor',[0 ...
    0.4470 0.7410], "LineStyle","--")
hold on
rectangle('Position',[(Rad_RotorShaft+Clearance) h_Actuator_R w_RotaryPlate ...
    h_RotaryPlate],"linewidth",2,'EdgeColor',[0 0.4470 0.7410], "LineStyle","-")
hold on
rectangle('Position',[-(r_Actuator_Inner+w_Actuator) h_Actuator_R w_RotaryPlate ...
    h_RotaryPlate],"linewidth",2,'EdgeColor',[0 0.4470 0.7410], "LineStyle","-")
hold on

% rotary plate
rectangle('Position',[-(r_Actuator_Inner+w_Actuator) ...
    h_Actuator_R+h_StationaryPlate 2*r_RotaryPlate_Outer ...
    h_RotaryPlate],"linewidth",2,'EdgeColor',[0.8500 0.3250 0.0980], ...
    "LineStyle","--")
hold on
rectangle('Position',[(Rad_RotorShaft+Clearance) h_Actuator_R+h_StationaryPlate ...
    w_RotaryPlate h_RotaryPlate],"linewidth",2,'EdgeColor',[0.8500 0.3250 ...
    0.0980], "LineStyle","-")
hold on
rectangle('Position',[-(r_Actuator_Inner+w_Actuator) ...
    h_Actuator_R+h_StationaryPlate w_RotaryPlate ...
    h_RotaryPlate],"linewidth",2,'EdgeColor',[0.8500 0.3250 0.0980], ...
    "LineStyle","-")
hold on

% bearing R
b_R =rectangle('Position',[r_RotaryPlate_Inner+Clearance ...
    h_Actuator_R+h_StationaryPlate w_Bearing h_Bearing])
hold on
b_R.FaceColor = [.7 .7 .7];
b_R.EdgeColor = 'k';
b_R.LineWidth = 2;
% bearing L
b_L=rectangle('Position',[-r_RotaryPlate_Inner-w_Bearing-Clearance ...
    h_Actuator_R+h_StationaryPlate w_Bearing h_Bearing])
hold on
b_L.FaceColor = [.7 .7 .7];
b_L.EdgeColor = 'k';
b_L.LineWidth = 2;

% hub
rectangle('Position',[-(0.5*w_Hub) h_RotorShaft w_Hub ...
    h_Hub],'Curvature',0.4,"linewidth",2,'EdgeColor',[0.8500 0.3250 0.0980], ...
    "LineStyle","-")
hold on
plot(Blade_x_Bot, Blade_y_Bot, "-", "linewidth",2,'Color',[0.9290 0.6940 0.1250])
hold on
plot(Blade_x_Top, Blade_y_Top, "-", "linewidth",2,'Color',[0.9290 0.6940 0.1250])
hold on
plot(-Blade_x_Bot, Blade_y_Bot, "-", "linewidth",2,'Color',[0.9290 0.6940 0.1250])
hold on
plot(-Blade_x_Top, Blade_y_Top, "-", "linewidth",2,'Color',[0.9290 0.6940 0.1250])
hold on

% pitchlinks
plot(Pitchlink_x, Pitchlink_y, "ko-", "linewidth",4)
hold on
```

```
plot(-Pitchlink_x, Pitchlink_y, "ko-", "linewidth",4)
hold on
plot(Axis_x, Axis_y, "k", "linewidth",1, "LineStyle","-.")
hold on

title({'Rotor Hub and Swashplate Diagram'},'FontSize',tsize)
ylabel("Distance from Masttop, h / mm",'FontSize',tsize)
xlabel("Distance from Rotor Axis, r / mm",'FontSize',tsize)
ylim([-h_Mast h_RotorShaft+150])
xlim([-150 150])
axis equal
grid on

Stationaryline = line(NaN,NaN,'LineWidth',2,'LineStyle','-','Color',[0 0.4470 ...
    0.7410]);
Rotary = line(NaN,NaN,'LineWidth',2,'LineStyle','-','Color',[0.8500 0.3250 0.0980]);
Blades = line(NaN,NaN,'LineWidth',2,'LineStyle','-','Color',[0.9290 0.6940 0.1250]);
Actuators = line(NaN,NaN,'LineWidth',2,'LineStyle','-','Color',[0.4660 0.6740 ...
    0.1880]);

legend([Stationaryline Rotary Blades Actuators],{'Stationary (With ...
    Synchropter)','Rotary (With Head)','Blades','Actuators (Neutral ...
    Position)'},'Location',"northeast",'FontSize',tsize)

text(-210, 0.93*h_Swashplate_ASM,{'Rotary Swashplate',' with Bearings ...
    \rightarrow'},'FontSize',tsize)
text(-195, 1.3*h_Swashplate_ASM,'Pitch Links \rightarrow','FontSize',tsize)
%text(0.28*TotalLength_Left, 0.57*RPM_Engine,'Axel')
%text(0.50*TotalLength_Left, 0.2*RPM_Engine,'Secondary Shaft')
%text(0.82*TotalLength_Left, 1.9*Desired_Rotor_Revs,'Rotor Shaft')

hold off

%% Other Mass Methods - UNUSED

%Tishchenko's Soviet Weight Formulaes Soviet Weight Formulae
```